

Master's Programme in Advanced Energy Solutions

Numerical Investigation of Diesel Spray Formation in a Marine Engine and Prediction of Liquid Fuel Approaching Cylinder Walls

Muhammad Ali Syed

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Author Muhammad Ali Syed		
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Thesis supervisor Prof. Ossi Kaario		
Thesis advisor(s) Dr. Zeeshan Ahmad, Dr. Jari Hyvönen		
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Abstract

Engine lubrication oil plays a critical role in ensuring smooth operations and preventing engine failures by reducing mechanical friction, forming protective films, preventing surface corrosion, sealing gaps, and maintaining component cleanliness. However, overtime, the degradation of lube oil quality can impact engine performance and shorten its operational life. In RCCI engines, the use of early pilot injection strategy can cause significant spray-wall impingement, increasing the chances of liquid-fuel deposition on cylinder walls. This creates local regions of unburned fuel near cylinder wall, which may seep past the piston rings into the oil sump. Such fuel dilution alters lubrication oil viscosity and accelerates its degradation, reducing its quality sooner than recommended interval. In this context, the thesis work aims to evaluate the scope of pilot-liquid fuel approaching cylinder walls by calibrating and validating GT-POWER based 1D spray and combustion predictive model, developed collaboratively by Wärtsilä Finland Oy and University of Naples Federico II, against range of experimental test data.

Various tuning constants from spray mass diffusion and flame propagation sub-models are used to calibrate the predictive model and to validate the experimental data sets taken from spray and engine performance test results. The validated model is used to evaluate zonal liquid fuel mass approaching cylinder walls and compare cases of two different umbrella angle specific nozzles.

The model effectively predicts the amount of liquid fuel in the last discretised zone adjacent to cylinder walls. The results indicate that with the increase in engine load the fuel spray vaporization accelerates and mass of liquid fuel near the cylinder wall decreases. While the comparison of umbrella angle sweeps indicate that larger spray angles cause significant amount of spray impinging on the walls as compared to smaller spray angles. Therefore, in RCCI engines, pilot-fuel nozzles at smaller spray angles provide greater possibility to avoid spray-wall impingement, lube oil dilution, and higher hydrocarbon emissions.

Keywords UniNa, RCCI, Spray-Wall Impingement, 1D Diesel Jet
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Preface and Acknowledgements

This master's thesis was conducted for Wärtsilä Finland Oy. In this work, W25 single cylinder research engine model was calibrated and validated using GT-POWER to predict engine performance and to evaluate the amount of fuel near the engine cylinder walls. The thesis was carried out to serve as a foundation to extend and validate the wall-film model to evaluate the amount of fuel seeping past the piston into the lube oil reservoir.

I am extremely grateful to Jari Hyvönen for giving me the opportunity to do my thesis work in his team. I am also grateful to my advisors Zeeshan Ahmad, Ben Smulter and Alberto Cafari for their extra ordinary guidance towards completing this thesis work. I want to thank Professor Ossi Kaario, who gave me good advice in structuring and polishing my thesis writing. I want to thank you for all courses you taught at Aalto University related to spray and combustion processes which truly increased my interest in working with internal combustion engines. I am also grateful to Alfredo Lanotte for his invaluable assistance in getting my GT-POWER model running and for helping me understand how to interpret the simulation results.

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Symbols and abbreviations

Symbols

η_{th}	Thermal efficiency
γ	Cutt off ratio
κ	Adiabatic Index
θ	Thermal efficiency
t_b	Cutt off ratio
ρ_l	Thermal efficiency
ρ_a	Density of air
$\bar{x}_{32}$	Sauter mean diameter
T_W	Wall temperature
T_{PA}	Pure adhesion temperature
T_{PR}	Pure rebound temperature
T_N	Nakayama temperature
z	Perpendicular distance from wall
d_e	Thickness of the boundary layer at maximum velocity
V_r	Velocity at co-ordinate z
V_{re}	Velocity at co-ordinate d_e
La	Laplace number
We	Weber Number
r_s	Surface roughness
u'_d	Tangential rebound velocity
v'_d	Normal rebound velocity
u_d^0	Tangential incident velocity
v_d^0	Normal incident velocity
θ_I	Incident angle
$E_{k,s}$	Splashing energy
E_k	Kinetic energy
c_f	Co-efficient of wall friction

α	Free jet impinging angle
$Z_{1/2}$	Half zone thickness
d_{bv}	Outer boundary layer
$\dot{m}_f$	Fuel flow rate
Y_{fav}	Average fuel fraction
T_l	Temperature of the liquid film
a_l	Thermal diffusivity
$\dot{q}_{in}$	Gas/film heat flux
$T_{l,m}$	Mean film temperature
$T_{l,s}$	Film surface temperature
h_t	Convective heat transfer coefficient
H_l	Latent heat of vaporization
δ	Liquid film thickness
T_∞	Ambient temperature
Φ	Total vaporization rate
$B_{M,i}$	Spalding mass transfer number of species i
$h_{y,i}$	Mass transfer coefficient
h_t	Heat transfer coefficient
λ_l	Thermal conductivity
N_l	Total species number in the liquid phase
$c_{p,g}$	Specific heat at constant pressure
Le_g	Lewis number
Sc_g	Schmidt number
Pr_g	Prandtl number
Y	Mass fraction
C_{ent}	Entrainment Rate Multiplier
C_{ign}	Ignition Delay Multiplier
C_{pm}	Premixed Combustion Rate Multiplier
C_{df}	Diffusion Combustion Rate Multiplier

Abbreviations

RCCI	Reactivity-Controlled Compression Ignition
OSCC	Optical Spray Combustion Chamber
CI	Compression Ignition
CR	Compression Ratio
SOI	Start of Injection
EOI	End of Injection
LP	Low Pressure
HP	High Pressure
LTC	Low Temperature Compression
HCCI	Homogeneous Charge Compression Ignition
EGR	Exhaust Gas Recirculation
IVC	Intake Valve Closing
TPA	Three Pressure Analysis
LFO	Light Fuel Oil
TDC	Top Dead Centre
BDC	Bottom Dead Centre

1 Introduction

Diesel engine was first developed by a German engineer Rudolf Diesel in 1892 [1], which uses the principle of compression ignition to burn fuel and produce power. Diesel engines draw air into the cylinder through intake manifold which is then compressed to a high pressure and temperature inside the cylinder. The fuel which is light fuel oil (LFO) or diesel in most cases is injected into the cylinder during the end of the compression stroke near Top Dead Centre (TDC) where the high temperature of the compressed air causes the fuel to ignite instantaneously causing a rapid expansion of the burnt gases and moves the piston downwards producing the power stroke. Diesel engines are considered as a feasible choice for marine applications such as for powering large marine vessels for its high thermal efficiency [2]. The diesel engines can operate at higher compression ratios as compared to gasoline engines and can convert large amount of fuel energy into useful work. Furthermore, diesel fuel offers greater amount of safety due to its low volatility, so it can be handled easily and can be stored in large quantities [3].

Despite the importance of diesel engines in marine sector, the primary concern is the high number of emissions diesel engines can produce. Marine engines are capable of operating on a wide range of fuel grades, and the use of low-quality fuels often cause high level of exhaust emissions [2]. Approximately, 20 million tonnes of nitrogen oxides (NO_x), 10 million tonnes of sulphur oxides (SO_x) and 1 million tonnes of particulate matter is emitted by the marine engines every year which contributes to 15% of NO_x, 4-9% of SO_x and 2.7% of CO₂ on global pollution, respectively [4, 5]. The International Maritime Organization implemented regulations to reduce emissions from shipping by adopting MARPOL, an International Convention for the Prevention of Pollution from ships [6]. The ANNEX VI is the latest regulation implemented under this convention in 2005 which set limits on SO_x, and NO_x emissions and prohibits the emissions of ozone depleting substances from ship exhausts [6]. Moreover, due to stricter emission legislations, high after-treatment cost and higher demands of crude oil owing to increased shipping activities, the dual fuel (DF) engines have gained attraction of engine developers as a suitable solution for shipowners.

Dual fuel engines work on the lean burn combustion concept wherein low reactivity fuel such as natural gas is premixed with the intake air before the intake valves open and a high reactivity fuel such as LFO or diesel works as a pilot fuel and burns the compressed premixed fuel-air mixture inside the combustion chamber [7]. DF engines provide efficiency and brake mean effective pressure just like diesel engines with much reduced NO_x, SO_x, and particulate matter emissions [7]. Natural gas is an attractive fuel to be used as a primary fuel for these DF engines due to its environmental benefits as compared to conventional fuels and its availability in large quantities offer competitive pricing.

Irrespective of lower carbon content and higher heating value of natural gas as compared to other fuels, using it as a primary fuel in DF engines have a primary disadvantage i.e. methane slip [8]. Methane slip is an energy waste and second biggest contributor to global warming after CO₂ [8]. Its global warming potential is 86 which is 35 times higher than CO₂ on 20- and 100-year time interval scale [9]. Reactivity controlled compression ignition (RCCI) is a promising combustion technology to address this limitation of DF technology and further improve thermal efficiency [10]. In RCCI, the in-cylinder reactivity stratification is essentially controlled by using two fuels with different cetane number. By adjusting the ratio of these two fuels, employing multiple injections, using appropriate EGR rates, and controlling combustion phasing, the pressure rise inside the cylinder can be effectively optimized thus giving higher thermal efficiency and lower methane slip and NO_x emissions.

1.1 Research Problem

The early injection strategies for pilot fuel in RCCI type combustion engines are used to manage cylinder peak pressures during combustion. These injection strategies cause significant spray impingement on cylinder walls [11], which creates unburned fuel region near the cylinder wall and produces higher amount of HC, CO and soot emissions [12]. The unburned fuel that deposits on the wall seeps past piston rings into oil sump which alters the viscosity of the lubricating oil and hence affects engine life [13]. The purpose of this research is to find the amount of liquid fuel approaching the cylinder walls by calibrating and validating the spray jet and combustion models against the data extracted from the experimental campaigns for Wärtsilä W25 single-cylinder research engine fuelled with a LFO as pilot fuel and Natural Gas as the main fuel.

The spray jet and combustion models used in this thesis are developed for W25 single cylinder research engine under a collaboration between University of Naples Federico II, Italy and Wärtsilä which will be referred as UniNa model hereafter in this thesis work. The UniNa model is an externally developed 0D/1D engine simulation framework in GT-POWER, which incorporates detailed sub-models for flame propagation, spray dynamics, combustion processes, emission formation and other relevant phenomenon. This external framework is integrated with GT-POWER for comprehensive analysis of various engine configurations including advance combustion concepts such as Reactivity Controlled Combustion Ignition (RCCI).

1.2 Objectives

The following objectives are aimed to achieve during this master's thesis:

1. Literature survey to gain understanding of various topics related to injection strategies in diesel engines, diesel spray formation, its characteristics and fuel-wall interaction.
2. Validation of the UniNa GT POWER model for the W25 RCCI-type combustion case using LFO as pilot fuel.
3. Validation of the UniNa spray characteristics with OSCC optical data, particularly spray penetration, head velocity and cone angle.
4. Evaluation of the last zone fuel fraction adjacent to the cylinder liner for several experimental data sets.
5. Comparison of cases for spray angle variations of the W25 pilot injection.

1.3 Limitations

For calibration of engine performance, Gamma Technologies recommends using at least 25 operating points which are spread over the whole engine speed and load map. But due to time constraints only 5 test points covering low to high loads with constant engine speeds are used for calibration of spray and combustion models.

Also, the time constraint has limited the number of test points to 4 for validation of the models due to behaviour of 5th test point in the DI combustion mode at full load.

1.4 Contents of the thesis

The thesis content is distributed in such a way that the first part gives detailed literature review for the research problem being studied, followed by the methodology used to develop the procedure to calibrate the GT-POWER model against the experiments performed on single cylinder research engine and optical spray chamber. The third part discusses the validation of results and prediction of liquid fuel mass approaching the cylinder walls. Finally, the last part concludes the thesis work briefly and presents detailed recommendations for the future work to be undertaken following the completion of this research work.

2 Literature review

2.1 Diesel Engine Theory

Like gasoline engines, diesel engines are also designed to work either on 2-stroke or 4-stroke cycle [1]. The 2-stroke engine produce power in one complete revolution of the engine crankshaft. The engine is designed so that the four processes such as Induction, Compression, Power and Exhaust occur simultaneously between 2-strokes of the engine cylinder. The 2-stroke engine was developed by a Scottish engineer Dugald Clerk with his dual cylinder configuration released in 1880 [1]. The design was further improved by an Englishman Joseph Day who rendered it with a valveless design. The first 2-stroke engine was practically introduced by an engineer Alfred Angus Scott from Yorkshire, England where he used a twin-cylinder water-cooled design for motorcycles [1]. On the other hand, a 4-stroke produce power in two complete revolutions of the crankshaft. The engine is designed such that all four processes of Induction, Compression, Power and Exhaust occur separately in four distinct strokes. This engine was first designed by a German engineer Nicolaus Otto, and he patented his design in 1877 by further improving his own design [1]. Diesel engine can also be classified in other ways such as based on air charging method, fuel injection method, rated crankshaft rotation speed and by cooling medium they use etc. Diesel engines can be naturally aspirated, mechanical supercharged or aerodynamically turbocharged engines with direct or indirect injection of fuel.

Diesel engine with rated crankshaft rotation of approximately 300 rpm is classified as low speed, between 300-1000 rpm as medium speed and faster than 1000 rpm as high-speed diesel engines [14]. Due to high compression ratios and no limitation on autoignition, diesel engines give high thermodynamic cyclic efficiency, high power, and lower levels of fuel consumption and CO₂ emissions as compared to gasoline engines. Equations (1) and (2) shows that the thermal efficiency of gasoline and diesel engines. Their comparison shows that for same compression ratio gasoline engines have higher thermal efficiency as compared to diesel engines but in practice higher compression ratios can be achieved for diesel engines which make them more thermally efficient [15].

$$\eta_{th} = 1 - \frac{1}{CR^{\kappa-1}} \quad (1)$$

$$\eta_{th} = 1 - \frac{1}{CR^{\kappa-1}} \frac{\alpha^{\kappa} - 1}{\kappa(\alpha - 1)} \quad (2)$$

In the above equations, η_{th} is the thermal efficiency, CR is the compression ratio, α is the cut-off ratio, κ is the ratio between specific constants of pressure and volume respectively.

2.1.1 Diesel Combustion Basics

The combustion in diesel engine consists of four main stages which are described in Figure 1. These four stages are i) ignition delay, ii) premixed burning, iii) mixing controlled combustion and, iv) late burning/ combustion [1]. These phases are explained below:

i. Ignition Delay

It is defined as the time taken by the combustion process to start after the fuel is injected into the combustion chamber. It can be further classified into following two parts:

- Physical Delay - This delay is caused by the physical processes involved in the liquid fuel atomization, vaporization and mixing of these fuel vapor with the entrained air.

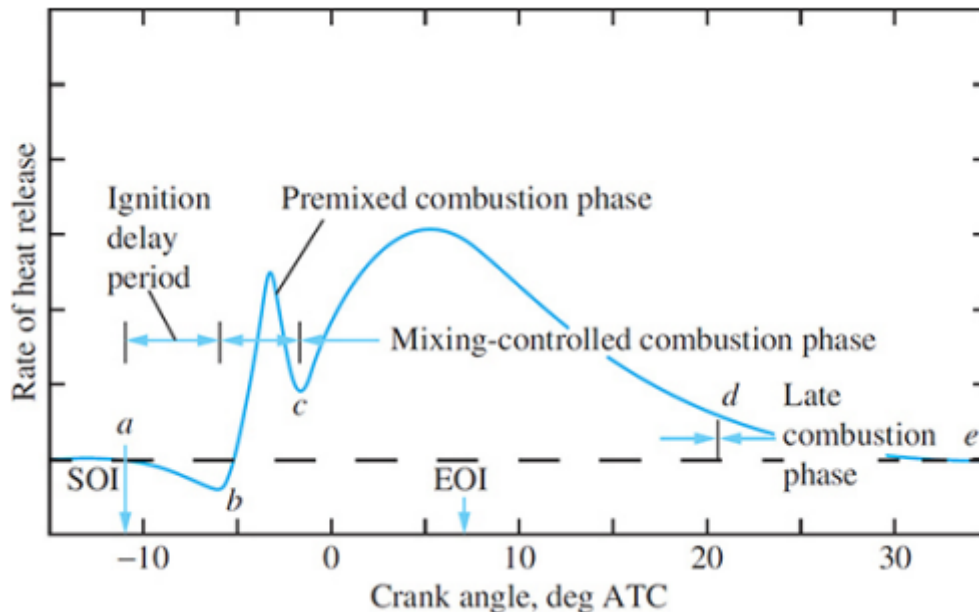


Figure 1: Combustion process of Diesel Engine [1]

- Chemical Delay - This delay is caused due to the time taken for achieving the chemical conditions for autoignition. Here preliminary heat release combustion reactions occur among fuel, air and residual gas mixture that eventually leads to autoignition process.
- ii. Premixed burning
- This phase is also known as rapid combustion phase, wherein the fuel mixed with the air during the delay period causes rapid pressure rise.

The high heat-release rate that result during this phase comes from the addition of burning mixture to the fuel which in turn burns it.

iii. **Mixing Controlled Combustion Phase**

In this phase the rate of burning is mainly controlled by the fuel vapor-air mixing process. During the previous phase the temperature and pressure rises to an extent where any fuel injected in this region burns instantaneously.

iv. **Late Burning/ Combustion**

In this phase a post injection strategy is employed during the expansion stroke, where unburned fuel fraction of fuel energy that exist in the soot is ignited to increase air/fuel mixing and hence the heat continues to release at a decaying rate.

2.1.2 Diesel Fuel Injection System

Fuel injection system in any engine is used to deliver required amount of fuel into the engine cylinders, while precisely controlling the parameters mainly injection timing, pressure and atomization. Most common types of fuel injection systems are pump-line-nozzle, unit injector, and common rail type injection systems. The injection system usually consists of a low-pressure and a high-pressure system. The LP system includes fuel tank, a fuel pump, filter and low-pressure lines while the HP system includes the high-pressure lines and the injectors (nozzles) [16]. A schematic diagram of a diesel fuel injection system is shown in Figure 2.

The injection system mostly used in marine engines is the common rail type injection system with solenoid valve injector [17]. The fuel which is diesel in most cases is introduced into the injector through an inlet passage to the injection nozzle and an inlet throttle to the valve control chamber which is connected to a return passage through outlet throttle which functions with a solenoid valve. Equal rail pressures on the valve control chamber and nozzle chamber keep the nozzle needle in closed position against the opening pressure on the valve piston's pressure shoulder. The solenoid valve spring also keeps the valve ball seated into outlet throttle exit passage [18].

When the solenoid valve is energized with a current the valve spring elastic force increases which lifts the valve ball from the outlet throttle seat. This causes the fuel to flow into the fuel tank through the return passage and decreases the pressure in the valve control chamber as compared to the pressure in the nozzle chamber. The drop in pressure decreases the force on the control piston and lifts the nozzle needle from the seat and hence starts the injection. The amount of fuel injected depends on the operating time of the solenoid valve while the nozzle's needle opening speed is determined by the differential flow between the inlet and outlet throttle [18].

To stop the injection the solenoid valve is de-energized and the valve spring pushes the armature and causes the seat ball to press against the opening of the outlet throttle. This raises the pressure in the valve chamber and equals to the pressure in the nozzle chamber. The force from the valve piston increases as compared to the upward force on the needle and forces the needle to rest on its seat. This process stops the injection from the injector [18].

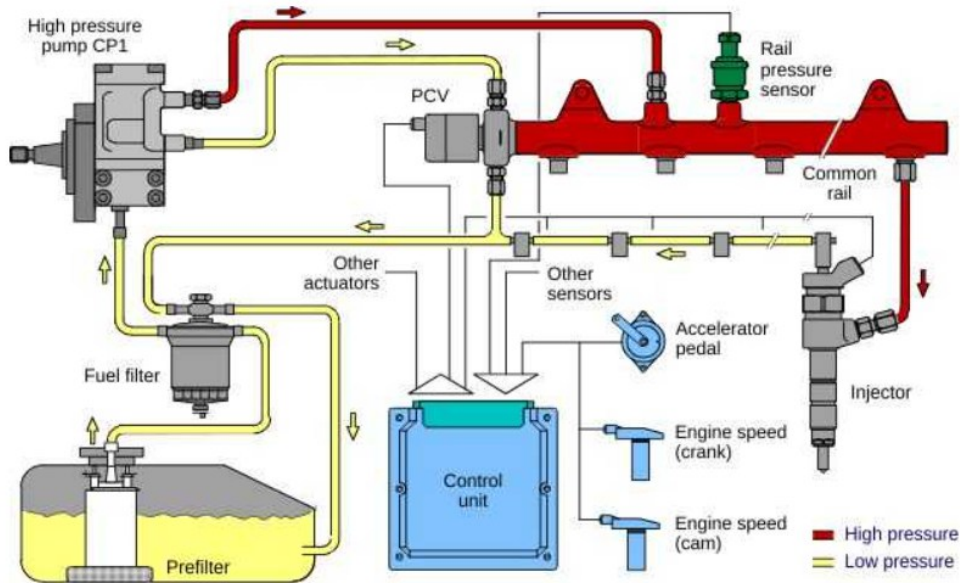


Figure 2: High pressure rail injection system [16]

2.1.3 Fuel Atomization and Spray Formation in Diesel Engines

Fuel atomization and hence spray formation are of great importance in a diesel engine for efficient combustion where a large number of fuel droplets are required to enhance vaporization, improve fuel-air mixing and increase combustion efficiency. For convenience the term atomization here refers to the physical disintegration of the continuous liquid jet or liquid core into number of smaller droplets and ligaments. The atomization of the liquid fuel is initiated with a primary breakup followed by fine droplets formation during secondary breakup.

2.1.3.1 Primary Breakup

The primary breakup of the fuel is initiated as the liquid jet flows out of the nozzle wherein the liquid core starts to breakup due to the occurrence of following flow regimes at the exit of the injection hole; i) laminar flow, ii) turbulent flow and iii) cavitating flow. Various reasons make the liquid jet to disintegrate into droplets. The main factor is the injection velocity which together with liquid properties and ambient conditions determines the atomization process of the liquid core. In the laminar regime the atomization

process is carried out by aerodynamic-induced atomization while in the turbulent regime, variation of aerodynamic forces does not affect the atomization process. In addition, under the practical injection pressures the flow in the injection holes is observed to be cavitating two-phase flow [19]. The following three atomization models can identify the effect of atomization process on the spray characteristics of the fuel:

- i. **Aerodynamics-Induced Atomization** - This model is based on Kelvin-Helmholtz instability phenomenon wherein waves start to develop on the surface of the liquid jet as it flows out of the injector. This is caused due to the relative motion of the injected fuel with gas inside the combustion chamber. The growth rate of these waves and the disintegration of the liquid jet into droplets is determined through a dimensionless parameter known as Weber Number which is defined as the ratio of inertial forces to surface tension forces. The effect of liquid viscosity on the breakup of liquid jet can be accounted through another dimensionless number known as Ohnesorge number. Ohnesorge number is defined by the ratio of internal viscosity dissipation forces to the surface tension energy which is used to describe the tendency of a liquid drop to stay together with the liquid or to fly apart [19].
- ii. **Jet Turbulence-Induced Atomization** - The flow conditions where the jet is in the fully turbulent regime, the disruption of the liquid surface film is caused by the radial velocity component in the jet, followed by the primary breakup of the liquid jet. This type of jet if even injected into vacuum can disintegrate due to its own turbulence characteristics [19]. A model given in [20] reported that the turbulent fluctuations present in the liquid jet emerging from injection hole produce initial surface disruption, which occur at a certain characteristic wavelength that is proportional to the correlation length of turbulence.
- iii. **Cavitation-Induced Atomization** - In this atomization process, cavitation initiated at the exit of injector holes cause the liquid jet exiting the nozzle holes to disintegrate into smaller droplets due to collapsing of these cavitation bubbles. The pressure around the jet exiting the nozzle is greater as compared to the pressure inside the cavitation bubbles which causes them to gradually collapse as the jet is accelerated outwards due to the turbulence inside the injector. This process of collapsing causes disruptions on the jet surface until they start to disintegrate into smaller droplets as the bubbles reach the jet surface [19]. The cavitating flow and the disintegration of the liquid core exiting the injector holes can be observed in Figure 3 [21].

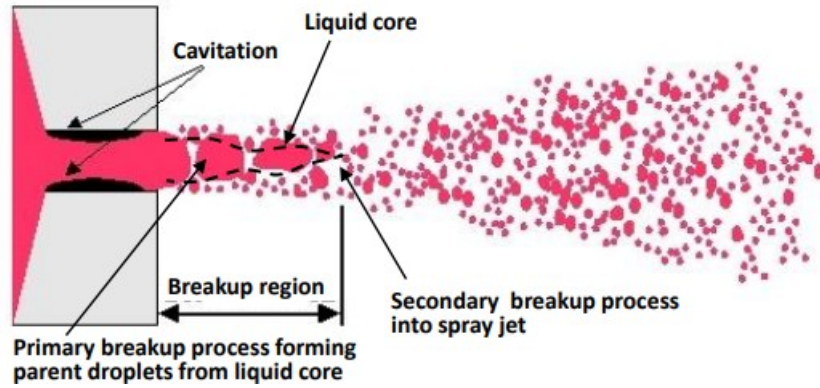


Figure 3: Break up model of diesel fuel jet core [21]

2.1.3.2 Secondary Break-up

The liquid jet that undergoes primary break-up further breaks into smaller droplets as they move into the surrounding gas. The relative velocity between the droplets and the surrounding gas develops non-uniform pressure distribution around the droplets causing them to undergo deformation. This process of droplets break-up is very intricate, and such processes largely depend on the surface tension, viscosity and inertia of liquid droplets under consideration [19].

The secondary breakup of these droplets is controlled through a dimensionless number known as Weber number which accounts for inertial and surface tension forces between the droplet and the surrounding gas [21]. The Weber number as discussed in previous section can be defined by the ratio of inertial force to the surface tension forces, and it can be expressed as shown in equation (3). The Figure 4 shows the secondary breakup of droplets which can be classified into three mechanisms based on droplets Weber number.

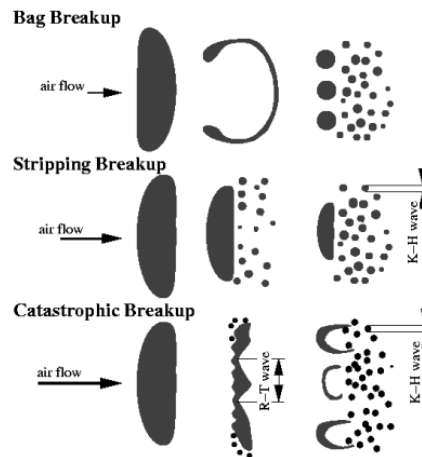


Figure 4: Schematics of droplets break-up [21]

Bag breakup mode occurs at a gas Weber number of $6 < We < 80$, wherein the droplets are seen to deform into a thin film having a bag shape. With the increase in the radius of the bag, the film thickness decreases further until hole forms and bursts the film into spray droplets [22]. The Stripping breakup or also called shear breakup occurs at a Weber number of $80 < We < 350$ while the Catastrophic or surface wave breakup occur at a Weber number of, $We > 350$ [21].

$$We = \frac{\rho_a V_{In}^2 d_l}{\sigma} \quad (3)$$

In the above equation, ρ_a represents the density of liquid droplet, V_{In} the normal incident velocity of droplet, d_l the droplet diameter and σ as the surface tension respectively.

2.1.4 Structure of Fuel Spray and its Characteristics

2.1.4.1 Spray Cone Angle

There are several definitions proposed by researchers regarding the spray cone angle of the fuel jet exiting the injection nozzle. According to Zama et al., spray cone angle is the angle θ defined by the maximum angle between the peripheries of the spray. He also defined the spray cone angle by the spray width W at a fixed axial location along the length of the spray where z is the axial length of spray at width W [23].

$$\theta = \tan^{-1} \left(\frac{W}{z} \right) \quad (4)$$

The most used empirical relation for spray cone angle evaluation is given by Hiroyasu et al., where they expressed spray cone angle by the following relation by analyzing various photographs of sprays [24]:

$$\theta = 83.5 \frac{L}{D} \times -0.22 \frac{D}{D_0} \times 0.15 \frac{\rho_a}{\rho_l} \times 0.26 \quad (5)$$

In the above equation, θ is the spray angle, L is the length of nozzle, D is the diameter of nozzle hole, ρ_a is air density and ρ_l is fuel density respectively.

2.1.4.2 Spray Tip Penetration

According to Payri et al., spray tip penetration is defined as the distance between the orifice exit and the furthest pixel achieved by the fuel spray in polar coordinates as shown below in Figure 5 [25].

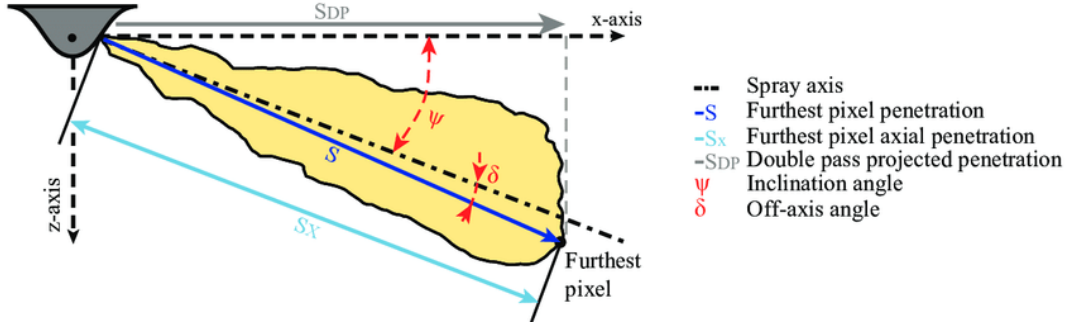


Figure 5: Definition of Spray tip penetration [25]

The injection speed and the spray penetration lengths have great effect on the air utilization inside the combustion chamber and fuel-air mixture. Hiroyasu et al. reported the effect of ambient gas pressures and temperature changes on the spray tip penetration. They observed that with the increase in ambient pressure the spray tip penetration decreases while it does not affect significantly with temperature changes even if the temperature is changed from room temperature to 590K [24]. They formulated the following equations for spray tip penetration by applying the jet disintegration theory given by Levich in [26]:

$0 < t < t_b$

$$S = 0.39 \left(\frac{2\Delta P}{\rho_l} \right)^{0.5} t \quad (6)$$

$t > t_b$

$$S = 2.95 \left(\frac{\Delta P}{\rho_a} \right)^{0.25} (Dt)^{0.5} \quad (7)$$

$$t_b = 28.65 \frac{\rho_l D}{(\rho_a \Delta P)^{0.5}} \quad (8)$$

In the above equations, S represents the spray penetration length, D is nozzle hole diameter, t_b is the breakup time, ρ_l is the fuel density, ρ_a is the air density and ΔP is the mean effective pressure drop across the nozzle respectively.

2.1.4.3 Spray Droplet Size Distribution

Fuel spray size distribution plays an important role in diesel engine combustion and its emission characteristics as it greatly affects the fuel-air mixing inside the combustion chamber. The smaller the size of the fuel droplets, the higher the vaporisation rate, resulting in fine droplet spray distribution. However, the fuel droplets may lose momentum with much smaller sizes, and the fuel-air mixing will no longer be therefore effective [21]. Several researchers investigated the spray size distribution through laser diagnostic techniques such as laser diffraction-based methods and proposed various

mathematical expressions in the literature. These results have shown that the outer zone of the spray contains larger size droplets as compared to inner zone near spray axis.

Takeuchi and Murayama et al. studied the droplet size distribution of diesel fuel in a quiescent atmosphere under room temperature and high pressures using three injector nozzles: throttle, pintle and hole-type. While observing the spray size distribution they observed that the wavy flow which appears in the secondary breakup of fuel disappears after twice the injection period while the atomization process of the fuel gets stabilized after three times the injection period. The maximum atomized droplet's diameter they observed in their study was 200, 165 and 130 micrometres for throttle, pintle and hole- type nozzles [27].

Another study was done by Li and Nishida et al. on spray droplet size distribution in Gasoline Direct Injection (GDI) engines under ambient pressures of 0.1 and 0.4 MPa using swirl type injectors. They observed the results via Mie scattering images and found wide cone spray structures at 0.1 MPa and narrow cone at 0.4MPa. The temporal variation of SMD (Sauter Mean Diameter) and DV90 (90% of total liquid volume where all small droplets lie) values for both ambient pressures have shown a decrease in their values from 1.5ms to 2.0ms after start of injection (ASOI) while from 2.0ms to 3.0ms ASOI the values show a decrease for 0.1MPa while an increase for 0.4MPa ambient pressure. The high air resistance during injection in 0.4MPa ambient conditions results in fine atomization at 1.5ms ASOI where the measured droplets just pass the spray leading edge but as the time passes until 3.0ms, where the measured droplets are in the middle of the spray stream the strong effects from the spray induced vortex-like air at the periphery of the spray moves smaller droplets into spray axis and these droplets move downward with the fast moving air leaving larger droplets at 3.0ms which increases both SMD and DV90 values [28]. The lowest SMD and DV90 values for 0.1MPa ambient condition were around 40 and 10 micrometres while for 0.4 MPa ambient conditions they were around 22 and 8 micrometres.

Takeuchi and Murayama et al. further used their microscopical photographs to find droplet size distribution around the spray axis. One such measured spatial distribution of droplets is shown in Figure 6 after start of injection timing (ASOI) of 6.1ms. The upper portion above spray axis show Sauter mean diameter while the lower portion below spray axis show arithmetic mean droplet diameter. The contour lines here represent the droplets with same diameters. The spatial distribution remains uniform at the start of injection, after which it becomes non-uniform until 1.5ms but at 6.1ms ASOI uniform distribution is observed according to Takeuchi and Murayama et al. [27]. Fine droplets can be observed from Figure 6 near injection nozzle periphery. They stay near exit of the injection nozzle because they lose momentum after they are generated from the jet-flow during injection of jet into the chamber and cannot travel further into the chamber due to lower velocities.

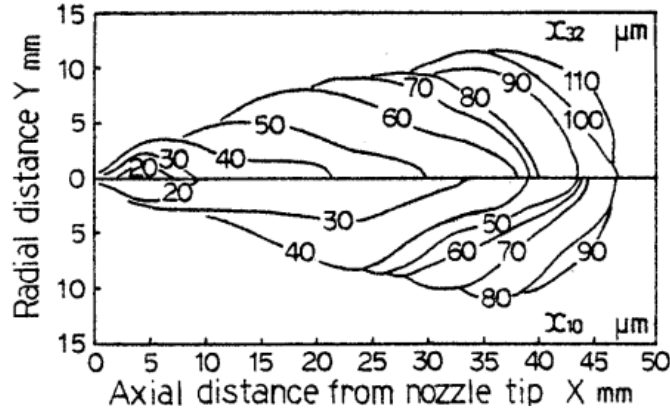


Figure 6: Spatial distribution of mean droplet diameter after 6.1ms ASOI

Takeuchi and Murayama et al. integrated the spatial distribution of droplets measured at all local regions and subsequently found the spatial distribution of droplets in the entire spray using the Nukiyama-Tanasawa distribution function as seen below:

$$\frac{dn}{dx} = nAx^\alpha \exp(-Bx^\beta) \quad (9)$$

where n represents total number of spray droplets, x is the drop diameter while indexes α is fixed at 2 and value of β describes uniformity of distribution of droplets. The value of this distribution function for the throttle nozzle can be found by using following expressions for constants and indexes in equation (9) [27]:

$$n = 150t^{7.87} \quad (10)$$

$$A = (-2.26t) \quad (11)$$

$$B = 0.003t^4 - 0.008t^3 - 0.291t^2 + 0.152t - 0.529 \quad (12)$$

$$\beta = x^{0.034^2 - 0.210t + 0.793} \quad (13)$$

2.1.5 Injection Strategies in Diesel Engines

Diesel engines that commonly employ high pressure common-rail injection systems offer higher amount of flexibility to tune injection timings and implementing multiple injection strategies, which can greatly reduce engine noise and emissions [29]. The injection strategies can be divided into pre, main and post injection strategies. Most of the researchers have employed pre and main injection strategies while others also used all three types of strategies in their research.

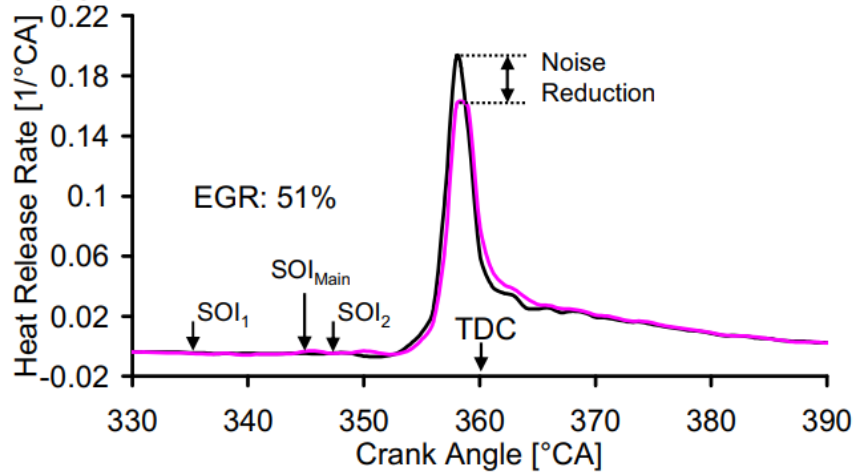


Figure 7: Comparison of single and 2 injection strategies [30]

Pilot injection or pre injection is mainly used to reduce the combustion noise of the engine which is primarily caused by the high-pressure gradient that occur by the rapid combustion of the premixed fuel-air mixture. This results in quieter engine performance as shown in Figure 7. The pre or pilot injection can also decrease the ignition delay period by rising the temperature and pressure inside the combustion chamber before the main injection of the fuel is started. A short ignition delay therefore reduces the premixed portion of the heat release [30].

Post-injection or after injection is a process where a small amount of fuel is injected after the main injection is completed. The purpose of the post injection is to increase air-fuel mixing and to raise the temperature of the exhaust gas. This process decreases the soot formation and helps to increase the EGR to reduce the NOx emissions [31]. The post injection introduces a small quantity of the fuel into the cylinder later in the expansion stroke after main injection, which reacts with the remaining air and releases heat in the exhaust [32]. The better use of the fuel during the combustion with post injection increases the combustion efficiency and reduces the overall hydrocarbon emissions. [33].

Asad et al. investigated the use of diesel low-temperature combustion (LTC) injection methods, specifically with single injection LTC and multi-injection LTC strategies in their research. The latter approach corresponds to the early homogenous charge compression ignition (HCCI) operation. Low temperature combustion or LTC is an efficient way to reduce NOx and soot emissions while maintaining high thermal efficiency. With heavy EGR at low load condition we can employ single shot injection close to TDC to enable LTC. This combination of high EGR rates with late injections close to TDC gives direct control over the engine performance by coupling combustion phasing and fuel injection processes. Therefore, high combustion efficiency can be achieved by optimizing the combustion phasing [30].

LTC can be enabled for mid-load conditions through early injection events where multiple injections are performed by difference of milliseconds before TDC. In this case heat release phasing cannot be controlled now directly by fuel injection. Instead, the combustion phasing which was controlled previously by the injection timing in the single shot LTC can now be only possibly controlled by the chemical kinetics during combustion and by adjusting charge temperatures [30].

2.1.6 Droplet Wall Interaction in Diesel Engines

Spray wall interaction in DI combustion engines is an important phenomenon as it effects the overall performance and emissions of the engine. The spray impingement on the walls has significant effect on the vaporisation of the fuel droplets and fuel-air mixture. The droplets that interact with the wall either stick or bounce back from the surface depending on wall characteristics such as roughness and temperature. The heat transfer from the hot walls to the spray cause them to vaporise at a higher rate which enhance the mixing process and hence increase engine performance [19].

Emission studies have shown that the spray impingement on the walls of the diesel engine produce significant number of unburned hydrocarbons during exhaust stroke due to the wall film formation on piston and cylinder walls [19]. The deposition of fuel droplets or rebounding from the wall depend on the inertial forces during their impact on the walls of the cylinder. The droplets experience rapid spreading and deformation, and this pressure is relieved as the droplets move along the cylinder wall. This movement of the flow can be stable or unstable based on the spreading flow conditions. In case of stable flow, the droplets will rebound from the surface if the incident kinetic energy of the droplets is higher than the sum of the adhesion and dissipated energies. While in case of unstable flow, the liquid droplets will disintegrate causing the release of secondary droplets and part of them get deposited on the walls [19].

Bai et al. identified different impingement regimes that an impinging droplet may undergo under various conditions. The schematic of these regimes can be seen in Figure 8 and are discussed in detail below [34]:

- i. **Stick** – In this regime the droplet adheres to the impinging wall in an almost spherical shape. This is due to the low impact energy of the droplet where the wall temperature T_w is below the characteristic temperature, T_{PA} known as pure adhesion temperature.
- ii. **Spread** – In this regime the droplet impinges with a moderate velocity on a dry or a wetted wall. If it impinges on a dry wall, it spreads out to form a thin film or on the other hand when it impinges on a wetted wall it merges with the pre-existed film.
- iii. **Rebound** – In this regime the droplet bounces back after impinging on the wall. This regime can occur in two cases; on a dry wall when T_w

$\geq T_{PR}$ (pure rebound temperature), where liquid droplet bounces off the hot wall surface due to intervening vapor film and in second case on a wetted wall, where the droplet impact energy is low and bounces off the wall due to intervening air film between the droplet and the liquid film.

- iv. **Rebound with breakup** – Here the droplet bounces back from a hot surface, $T_W \leq T_{PR}$, and breakup into 2 or 3 droplets.
- v. **Boiling-induced breakup** – In this regime the droplet disintegrates even at very low collision energy with a hot wall whose temperature lies near the Nakayama temperature T_N . It is the temperature at which the droplet achieves its maximum rate of evaporation.
- vi. **Break-up** – In this regime the droplets deform to form a radial film on the surface with high temperature, $T_W > T_{PA}$, the thermal degradation of the film disintegrates it in random manner.
- vii. **Splash** – In this regime the droplets collide with the wall with high impact energy forming a crown. The droplets become unstable at the periphery of this crown and disintegrate into small fragments.

Factors such as droplet size, droplet velocity, surface tension, viscosity, wall roughness, wall temperature and the presence of the wall film are primarily responsible for the hydrodynamic processes during droplet impingement on surfaces. While the dimensionless parameters that include Reynolds number, Weber Number and Ohnesorge number greatly influence the overall wall impingement characteristics of these droplets on cylinder walls.

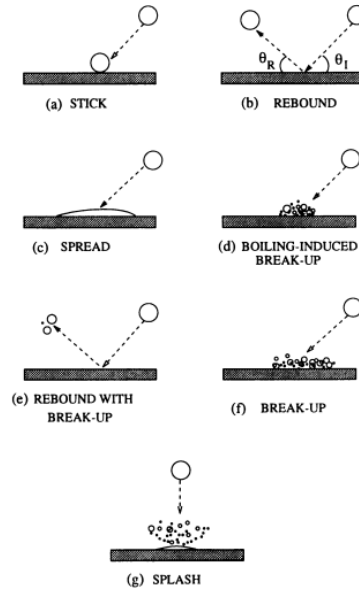


Figure 8: Schematic of impact regimes [34]

2.2 Modelling

This section briefly describes the one-dimensional models for diesel jets, spray-wall interaction, droplet evaporation and wall-film evaporation. The section further describes different combustion modelling systems available including predictive and non-predictive combustion models. In the last, the types of combustion models available in GT-POWER will be discussed in detail.

2.2.1 One-dimensional Diesel Jet Model

Musculus and Kattke developed a 1D diesel jet model to study the behaviour of sprays and entrainment waves for transient diesel jets. They observed in transient diesel jet measurements that the fuel in the wake region of the injected spray mixes more rapidly with ambient gases as compared to steady jets. This creates fuel-lean condition near injector after end-of-injection (EOI) which may not burn completely [35]. In the model described below, both the researchers studied the fuel mass entrainment behaviour in transient diesel jets using a simple 1D jet model.

They based their model on the analysis of steady injections using control volume done by Naber and Siebers. The latter have assumed a constant spreading angle for the jet with zero velocity slip with the entrained gases. They further assumed that injection velocity is constant and start instantly. They also simplified their analysis to one dimension by considering a uniform profile for axial velocity over the jet cross-section and derived an expression analytically for penetration function [35]. Musculus and Kattke found that this solution treats the velocity of jet as an invariable quantity w.r.t time at any axial location and cannot predict the dynamics of a jet having unsteady injection rate.

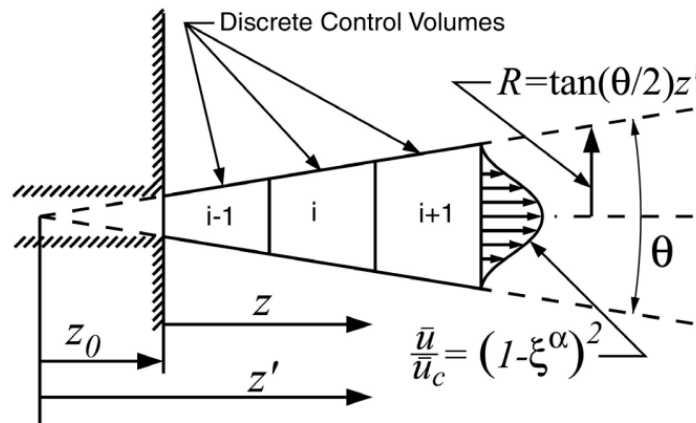


Figure 9: 1D discrete control volume description for transient diesel jet [35]

In their solution as shown in Figure 9, they discretized the whole domain into smaller control volumes having their own transient equations for jet axial momentum and fuel mass transport. To solve these transient transport terms, following seven assumptions were considered to expand Naber and Siebers steady-jet model [35]:

1. The fuel is treated as liquid. Consequently, the jet is considered non-vaporizing
2. The downstream jet velocity is slow, so compressibility has minor effect on diesel jet flow. Hence, the flow is considered incompressible.
3. The turbulent viscous forces can be neglected for each control volume as both the normal and shear forces are small due to small boundary velocities and their gradients.
4. For momentum transport equation only, axial convection will be considered while axial mixing due to momentum and turbulent diffusion are neglected.
5. For steady jets, the net force produced due to axial pressure gradient is negligible.
6. During and after end-of-injection (EOI) transient the jet spreading angle is assumed as constant.
7. During the end-of-injection (EOI) transient, the normalized radial profile as shown Figure 9 in remains unchanged.

Musculus and Kattke used these assumptions in each control volume to evaluate equations for transient transport of fuel mass (m_f) and axial momentum (M) as written below:

$$\frac{\delta m_f}{\delta t} = \dot{m}_{f,in} - \dot{m}_{f,out} \quad (14)$$

$$\frac{\delta M}{\delta t} = \dot{M}_{in} - \dot{M}_{out} \quad (15)$$

In the above equations, the over-dots express the integral fluxes of fuel mass and axial momentum crossing the upstream and downstream cross-sections of each control volume and their expressions can be expanded as follows:

$$\dot{m}_f = \rho_f \int \bar{X}_f \bar{u} dA \quad (16)$$

$$\dot{M} = \int \bar{\rho} (\bar{u})^2 dA \quad (17)$$

where ρ_f is the liquid fuel density, $\bar{u}$ is the turbulent mean velocity, $\bar{\rho}$ is the turbulent mean density and $\bar{X}_f$ is the turbulent mean fuel volume fraction. Here cross-sectional area of the jet is $A = \pi(\tan(\theta/2)z')^2$ which is the function of distance from the virtual jet origin z' . Musculus and Kattke evaluated the integral in equation (16) by assuming profile for $\bar{X}_f$ and $\bar{u}$ proposed by Abramovich in [36]:

$$\frac{\bar{X}_f}{\bar{X}_{f,c}} = (1 - \xi^\alpha)^2 \quad (18)$$

$$\frac{\bar{u}}{\bar{u}_c} = (1 - \xi^\alpha)^2 \quad (19)$$

In the above equations, the subscript c indicates values at the jet centreline while ξ is the ratio of radial coordinate 'r' and jet width $R = \tan(\theta/2)z'$ as shown in Figure 9. The superscript α is used as an adjustable parameter which shows the evolution of the profile from uniform shape ($\alpha=\infty$) at nozzle exit to fully developed shape ($\alpha=1.5$) at distance downstream.

The centreline quantities in equations (18) and (19) can be made to relate cross-sectionally averaged quantities of fuel volume fraction $\bar{\bar{X}}_f$ and velocity $\bar{u}$ as given in [35] by using dA in equation (16) as $dA = 2\pi R^2 \xi \cdot d\xi$ which transforms the integral as:

$$\dot{m}_f = \rho_f \beta \bar{\bar{X}}_f \bar{u} A \quad (20)$$

While the turbulent mean density $\bar{\rho}$ in equation (17), is treated as constant at its cross-sectionally averaged velocity $\bar{u}$, so the integral transforms as [35]:

$$\dot{M} = \bar{\rho} \beta \bar{u}^2 A \quad (21)$$

where factor β is constant and accounts for the fuel volume fraction and velocity profiles shape. It ranges from $\beta=1$ for uniform profile to $\beta=2$ for fully developed profile. It is evaluated as follows [35]:

$$\beta = \frac{6(\alpha + 1)(\alpha + 2)}{(3\alpha + 2)(2\alpha + 1)} \quad (22)$$

In equation (21), $\bar{\rho}$ is proportional to liquid fuel density and ambient density while values of $\bar{\bar{X}}_f$ and $\bar{u}$ can be found from fuel mass and axial momentum relations in each control volume. All these cross-sectionally averaged values are computed as follows [35]:

$$\bar{\rho} = \rho_f \bar{\bar{X}}_f + \rho_a (1 - \bar{\bar{X}}_f) \quad (23)$$

$$\bar{\bar{X}}_f = \frac{m_f / \rho_f}{A \Delta z} \quad (24)$$

$$\bar{u} = \frac{M}{\bar{\rho} A \Delta z} \quad (25)$$

In equations (24) and (25), volume $A\Delta z$ is the discretized control volume space and Δz is its axial width as shown in Figure 9. Musculus and Kattke used the above development of equations to discretize the equations of fuel mass and axial momentum in (14) and (15) using an upwind differencing scheme. These discretized equations are written as follows [35]:

$$m_{f,i}^{t+1} = m_{f,i}^t + \rho_f [(\beta \bar{X}_f \bar{u} A)_{i-1}^t - (\beta \bar{X}_f \bar{u} A)_i^t] \Delta t \quad (26)$$

$$M_i^{t+1} = M_i^t + [(\bar{\rho} \beta \bar{u}^2 A)_{i-1}^t - (\bar{\rho} \beta \bar{u}^2 A)_i^t] \Delta t \quad (27)$$

Musculus and Kattke predicted using above model that after the end-of-injection (EOI) an increase entrainment region is formed downstream which travels twice the speed of the initial jet propagation rate and this is responsible for rapid stagnation of fuel-air mixtures near nozzle exit and spatial shifts with increase soot oxidation after EOI.

2.2.2 Spray-Wall Modelling

In a 0-D framework, a multizone model as shown in Figure 10 can be used to describe the spray-wall interaction in RCCI simulation [37]. Here the in-cylinder volume is divided into axisymmetric and cylindrical multiple zones each treated as a separate chemical reactor. Further conical shaped zones representing spray regions are combined with these cylindrical zones. These spray regions are produced using a multi-hole injector and therefore helps in classifying the in-cylinder zone into “in-spray zones”, zones that are formed through intersection of spray cones with cylindrical zones shown as red in Figure 10, and “out-spray zones”, zones that form outside the in-spray zones. The temperature and compositions in each zone are assumed different and therefore each zone has different combustion characteristics. By considering a section in a zone ‘z’ far from the injector, a mass balance equation across the section gives the heat transfer rate and mass transfer rate to the out-spray zone $z_{\frac{I}{O}}$ and the adjacent in-spray zones $z_{z,\pm 1}$ as given below [37]:

$$\dot{Q}_{z,\frac{I}{O}} = A_{z,\frac{I}{O}} \lambda_t \left(\frac{T_z - T_{\frac{I}{O}}}{\Delta r_{z,\frac{I}{O}}} \right) \quad (28)$$

$$\dot{Q}_{z,z\pm 1} = A_{z,z\pm 1} \lambda_t \left(\frac{T_z - T_{z\pm 1}}{\Delta r_{z,z\pm 1}} \right) \quad (29)$$

$$\dot{m}_{z,\frac{I}{O},i} = A_{z,\frac{I}{O}} \lambda_t \left(\frac{x_{z,i} - x_{\frac{I}{O},i}}{\Delta r_{z,\frac{I}{O}}} \right) \pm \dot{m}_{out,z,i}^{spray} \quad (30)$$

$$\dot{m}_{z,z\pm 1} = A_{z,z\pm 1} \lambda_t \left(\frac{x_{z,i} - x_{z\pm 1,i}}{\Delta r_{z,z\pm 1}} \right) + \dot{m}_{z,z-1,i}^{spray} - \dot{m}_{z,z+1,i}^{spray} \quad (31)$$

The above equations can be extended to the last zone inside the cylinder where it interacts with the cylinder wall. As the spray jet impinges on the wall a part of it sticks on the wall while the rest of the spray bounces back and travels along the cylinder wall which can be referred as wall jet. The amount of the spray jet that sticks on the wall depends on various factors such as wall temperature, density of fuel, surface tension of liquid fuel and non-dimensional parameters such as Weber number, Laplace number etc.

Consider a conical section of the in-cylinder volume having multiple zones as shown in Figure 11 having a length L from the centre of the injector, we can find the quantity of mass of fuel left each zone in the considered in-cylinder volume and from the last zone 'z' after the spray impinges on the wall of the cylinder by using the following relation [38]:

$$\dot{m}_{out,z}^{spray} = \dot{m}_{z,z-1}^{spray} * K \left(\frac{x - X_0}{L - X_0} \right) \quad (32)$$

In the above equation, $\dot{m}_{out,z}^{spray}$ is the rate of mass flow out from zone 'z', $\dot{m}_{z,z-1}^{spray}$ is the rate of mass flow from zone 'z-1' to zone 'z', K is the fraction of fuel spray mass coming out from the zone 'z-1', x is the distance of a zone from injector and X_0 is the position at which the outcoming mass flow activates. If we know the value of $\dot{m}_{out,z}^{spray}$, we could use the following fuel mass equation to evaluate the amount of fuel deposited on the cylinder wall:

$$\frac{\partial m_f}{\partial t} = \dot{m}_{f,in} - \dot{m}_{f,ex} - \dot{m}_{out,z}^{spray} \quad (33)$$

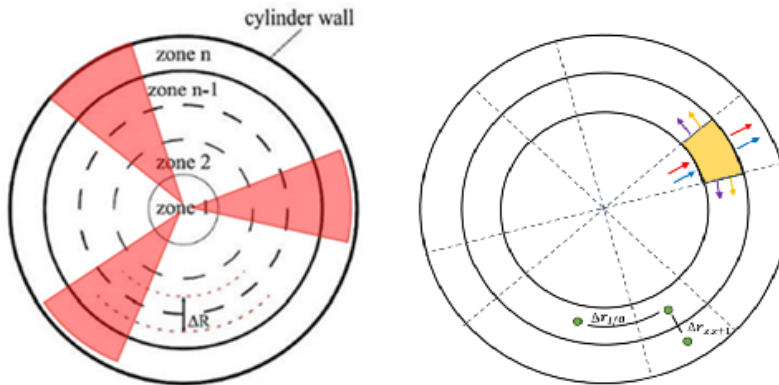


Figure 10: Cylinder volume divided in multiple zones

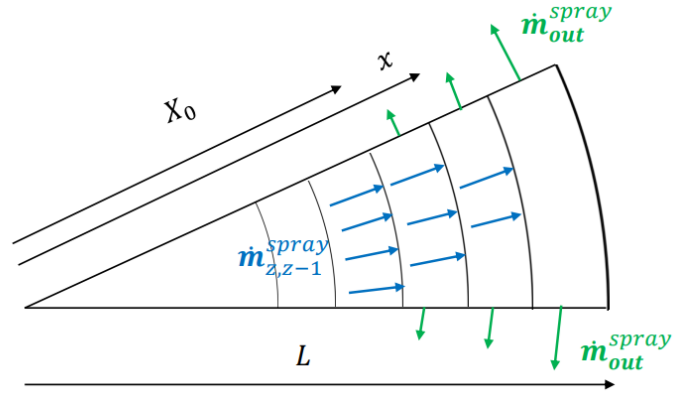


Figure 11: Conical section of the in-cylinder volume

2.2.3 Droplet Evaporation Model

A gas phase model developed by Abramzon et al. is used to analyse the mass and heat transfers near the droplet surface, and to find the instantaneous quantities such as vaporization rate ($\dot{m}$) and heat flux (Q_L) across the droplet surface [39]. The model solves the mass and energy balance equations for a single evaporating droplet to find relevant quantities mentioned above. The assumptions used for modelling droplet evaporation are listed as follows:

1. The gas phase heat and mass transfer are assumed to be in quasi-steady conditions.
2. There should be negligible pressure drop within the gas phase.
3. Film theory will be used to account for the effect of convection caused because of relative motion between the droplet motion and gas flow.
4. Fuel vapor distribution is assumed uniform over the droplet surface.
5. Temperature distribution is also assumed uniform with surface included.

As explained in assumption 3, the film theory is used to develop the concept of gas films with constant thickness (δ_T, δ_M) to take account for convective transport between droplet motion and gas flow. Thus, to model the resistance to mass and heat flow across droplet and gas flow, the thickness of these thermal (δ_{T0}) and diffusion (δ_{M0}) films for non-vaporizing spherical droplets is expressed as follows [39]:

$$\delta_{T0} = \frac{2r_s}{Nu_0 - 2} \quad (34)$$

$$\delta_{M0} = \frac{2r_s}{Sh_0 - 2} \quad (35)$$

In the above equations, r_s is the instantaneous droplet radius, Nu_0 is the Nusselt number, and Sh_0 is the Sherwood number respectively. Surface blowing of droplets increases the laminar boundary layer thickness which introduces the Stefan problem in evaluating these film thicknesses. To take account of Stefan problem, Abramzon and Sirignano introduced correction factors as follows [39]:

$$F_T = \frac{\delta_T}{\delta_{T0}}; F_M = \frac{\delta_M}{\delta_{M0}} \quad (36)$$

Abramzon and Sirignano considered another model problem where the laminar boundary layer flow over a vaporizing wedge. They assumed ranges of various parameters including B_T, B_M, β , Schmidt and Prandtl numbers. Here B_T and B_M are Spalding heat transfer and mass numbers which will be explained later and β is the wedge angle assumed in the range of 0 and 2π . Abramzon and Sirignano found that for an isothermal surface with constant physical properties of the problem fluid, we have a self-similar solution and these correction factors are independent of local Reynold Number. They are also insensitive to Schmidt and Prandtl non-dimensional numbers and variations in wedge angle, so we can approximate them as:

$$F_M = F(B_M), F_T = F(B_T) \quad (37)$$

In the above equation, $F(B)$ is the universal function and its value increases from 1 to 1.285 for values of B in the range of 0 and 8. This universal function can be written as:

$$F(B) = (1 + B)^{0.7} \frac{\ln(1 + B)}{B} \quad (38)$$

The film model is extended to assume uniform distribution of the heat and fuel vapor concentrations across the droplet surface. A mass balance is applied across the droplets with the extended film model yields the following expression for the instantaneous vaporization rate of droplets [39]:

$$\dot{m}_{drop} = \pi \bar{\rho}_g \bar{D}_g d_{drop} Sh^* \ln(1 + B_M) \quad p_{sat}(T_{drop}) < p_{cyl} \quad (39)$$

Or,

$$\dot{m}_{drop} = \pi \frac{\bar{k}_g}{\bar{c}_{p,fuel}} d_{drop} Nu^* \ln(1 + B_T) \quad p_{sat}(T_{drop}) = p_{cyl} \quad (40)$$

In the above equations, $\bar{\rho}_g$ is the average density, $\bar{D}_g$ is the binary diffusion coefficient, $\bar{k}_g$ is the thermal conductivity of the gas mixture in the film, $\bar{c}_{p,fuel}$ is the average vapor specific heat constant, B_M and B_T are Spalding mass and

heat transfer numbers, and Sh^* and Nu^* are non-dimensional parameters given as follows [39, 40]:

$$B_M = \frac{x_{fuel,drop} - x_{fuel,u}}{1 - x_{fuel,drop}}; B_T = \frac{\bar{c}_{p,fuel}(T_u - T_{drop})}{\Delta H_{vap}} \quad (41)$$

$$Sh^* = 2 + \frac{Sh_0 - 2}{F_M}; Nu^* = 2 + \frac{Nu_0 - 2}{F_T} \quad (42)$$

The equation (39) expressed above is used when the saturation pressure at droplet surface temperature is less than the actual or cylinder pressure and hence the governing phenomenon for vaporization will be due to mass transfer by diffusion while the equation (40) is used when both surface and cylinder pressures are equal, where the governing phenomenon for droplet vaporization is by heat transfer through convection.

In the equations of Spalding mass and heat transfer numbers, x_{fuel} is the fuel mass fraction, T_{drop} is the droplet surface temperature, T_u is the free stream flow temperature and ΔH_{vap} is the latent heat of vaporization while subscripts ‘drop’ and ‘u’ refer to droplet surface and external flow conditions. The non-dimensional parameters given in equation (42) should not be confused with actual Sherwood or Nusselt numbers which are non-dimensional mass and heat transfer coefficients.

Further, to find the heat penetrating across the droplet Abramzon and Sirignano used an energy balance equation across the droplet, to deduce the following heat transfer equation [39]:

$$\frac{dT_{drop}}{dt} = \frac{1}{m_{drop}c_{p,Fuel}} \left(\pi \bar{k}_g d_{drop} Nu^* \frac{\ln(1 + B_T)}{B_T} (T_u - T_{drop}) - \Delta H_{vap} \frac{dm_{drop}}{dt} \right) \quad (43)$$

In the above equation, the first term in the brackets represent the convective heat transfer from surrounding ambient while the second term represent the heat variation due to fuel vaporization. The first term in the equation leads to a temperature increase as in engine conditions the in-cylinder temperature is higher than the droplet temperature while the second term in the equation leads to temperature reduction.

2.2.4 Wall Film Evaporation Model

The model described below for wall film evaporation is a quasi-dimensional model. In this model, the distribution of temperature and component fraction for each species in the wall film is based on presumed profiles. This increases the computation accuracy as compared to one-dimensional models. Stanton and Rutland found in their research on multi-dimensional modelling of fuel-wall films that the liquid film thickness in engine relevant conditions

is less than 200 μm [41]. With such less film thickness, the conductivity can be assumed to be dominant only in the normal direction to the film, and the energy equation can be simplified as follows by neglecting the convective term owing to its smaller value as compared to diffusion term [40]:

$$\frac{\partial T_l(z)}{\partial \tau} = a_l \frac{\partial^2 T_l(z)}{\partial z^2} \quad (44)$$

where in the above equation T_l represents the liquid film temperature, z the distance from the impinging wall, a_l the thermal diffusivity and τ the time respectively. At the boundary of gas and liquid phases, the energy conservation follows the expression as:

$$h_t(T_\infty - T_{l,s}) - \lambda_l \left(\frac{\partial T}{\partial z} \right) \Big|_{z=\delta} = \dot{\phi} H_l \quad (45)$$

Above equation states that convective heat transfer from the gas to the liquid surface minus the conductive heat transfer through width of the liquid film is equal to the total vaporization rate times the latent heat of vaporization. Here h_t is represents convective heat transfer co-efficient, T_∞ and $T_{l,s}$ are ambient and surface temperatures of the liquid film, λ_l is the liquid thermal conductivity and δ is the thickness of the liquid film.

The thermal and mass diffusions within the liquid films greatly effects the vaporization of the fuel having multi-components. Moreover, the timescale of these diffusion processes is much comparable to the lifetime of the liquid films. So, Zhang and Jia et al. employed a high order polynomial expression to model the temperature and species fraction profiles within the liquid film shown in Figure 12. The temperature profile can be written in the following way [40]:

$$T_l(z) = a_0 + a_1 \left(\frac{z}{\delta} \right) + a_2 \left(\frac{z}{\delta} \right)^2 + a_3 \left(\frac{z}{\delta} \right)^A \quad (46)$$

The co-efficients in equation (46) are constants and are related to gas/film heat flux, wall temperature, mean film temperature and film surface temperature while the parameter A is the highest order of polynomial used in the equation. These constants can be computed by applying boundary conditions on the film shown in Figure 12 and for the sake of brevity the values of these constants are not described in this section.

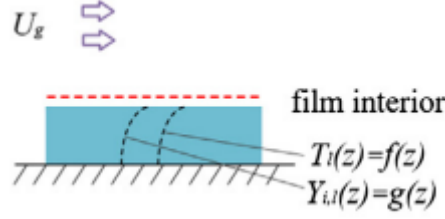


Figure 12: A quasi-dimensional model of wall film [40]

The expression for temperature profile and its constants can be substituted in the film energy equations (44) and (45) to find the derivative form of mean film and film surface temperatures as follows [40]:

$$\frac{dT_{l,m}}{d\tau} = \frac{a_l}{\delta^2} \frac{(A-3)\Omega + 2(A+3)T_{l,s} + 4AT_w - 6(A+1)T_{l,m}}{A-1} \quad (47)$$

$$\frac{dT_{l,s}}{d\tau} = \frac{a_l}{\delta^2} [(A+3)\Omega + 6(A+1)T_{l,m} - 2(2A+3)T_{l,s} - 2AT_w] \quad (48)$$

where $T_{l,m}$ is the mean liquid temperature, T_w is the wall temperature, $T_{l,s}$ is the liquid surface temperature and δ is film thickness.

Zhang and Jia et al. found that gas phase modelling primarily dominates the vaporization rate of the liquid film. Besides this, it is also responsible for higher amount of heat transfer between the liquid film and the surrounding environment. So, the total vaporization rate of the liquid film can therefore be determined by the following equation [40]:

$$\dot{\phi} = h_y \ln \left(\frac{\frac{\dot{\phi}_i}{\dot{\phi}} - Y_{i,v,\infty}}{\frac{\dot{\phi}_i}{\dot{\phi}} - Y_{i,v,s}} \right) = h_y \ln(1 + B_{M,i}) \quad (49)$$

where $\dot{\phi}$ is the product of density and velocity of gas mixture such that $\dot{\phi} = \rho_g V_g$. $B_{M,i}$ is Spalding mass transfer number of species i and $h_{y,i}$ is the mass transfer co-efficient which can be expressed in following forms:

$$B_{M,i} = \frac{Y_{i,v,s} - Y_{i,v,\infty}}{\frac{\dot{\phi}_i}{\dot{\phi}} - Y_{i,v,s}}; h_{y,i} = \left(\int_0^\infty \frac{1}{\rho D_{i,g}} \right)^{-1} \quad (50)$$

In the above equations, Y is the mass fraction of vapor, $D_{i,g}$ is the diffusion co-efficient and subscripts o is the initial value on axis z taken on the film surface, ∞ represents the extreme value along axis z away and normal to film surface, i represents the species and s represents surface here respectively.

The $B_{M,i}$ of species can be assumed equal to the B_M which is total Spalding mass transfer number then the total vaporization rate, $\dot{\phi}_i$ of the liquid film transforms into following form:

$$\dot{\phi}_i = h_y \ln(1 + B_M) \cdot \frac{\dot{\phi}_i}{\dot{\phi}} \quad (51)$$

where B_M , the total Spalding mass can be found through following expression:

$$B_M = \frac{\sum_{i=1}^{N_l} Y_{i,v,s} - \sum_{i=1}^{N_l} Y_{i,v,\infty}}{1 - \sum_{i=1}^{N_l} Y_{i,v,s}} \quad (52)$$

Zhang and Jia et al. used the Chilton-Colburn analogy to deduce an expression for the mass transfer co-efficient (h_y). This analogy develops a quantitative relationship between fluid flow and heat transfer quantities, within the gas boundary layer above the liquid film. The relation they deduced is as follows [40]:

$$h_y = \frac{h_t}{c_{p,g}} \times Le_g^{-\frac{2}{3}} \quad (53)$$

where $c_{p,g}$ is the specific heat of gas at constant pressure, Le_g is the Lewis number of gas and h_t is the convective heat transfer co-efficient which can be found in [40]. For sake of brevity, it is not explained here. Lewis number is an important parameter in wall film heat transfer can be defined by the following ratio of dimensionless numbers, Schmidt and Prandtl numbers:

$$Le_g = \frac{Sc_g}{Pr_g} \quad (54)$$

The Schmidt and Prandtl numbers are used compare the relative importance of momentum to mass and thermal diffusivity respectively. To model heat flux quantity from the gas phase to the liquid phase Zhang and Jia et al. used an energy balance equation to deduce the following expression [40]:

$$\dot{q}_{in} = h_t(T_\infty - T_s) - \dot{\phi}H_l \quad (55)$$

where H_l represents the latent heat of vaporization.

As explained in section 2.2.3, blowing or movable film surface due to high amount of mass transfer can resist the film heating and vaporization which also belongs to famous Stefan problem. A lot of researchers studied this problem in their research on multi-component droplet heating and evaporation. They concluded that the moving boundary has negligible effect on droplet

heating more common for smaller droplet. Due to this reason Zhang and Jia et al. neglected the effect of boundary layer movement on film heating and evaporation for the development of wall film evaporation model described above.

2.2.5 Combustion Modelling

Combustion models are used to predict the engine performance and emissions. They have seen an increased use in developing injection strategies, optimizing performance and reducing emissions of marine engines. So, it is important to have simulation tools available for engine design and calibration which combine accuracy and work with limited computational resources to test number of options for engine conditions. Phenomenological modelling techniques such as 0D and 1D thermodynamics combustion modelling seems to be a promising solution to model diesel combustion as compared to 3D CFD techniques which are more expensive and takes a lot of simulation time [35].

2.2.5.1 Thermodynamics based modelling

In this modelling technique, the whole cylinder is considered as one cell where the instantaneous combustion parameters such as pressure, temperature and rate of heat release etc. are estimated by applying mass, energy and state balance equations. The 0D thermodynamic models do not solve the spatial characteristics of the spray and therefore 1D phenomenological models seems a compromising solution to this problem [42].

2.2.5.2 Fluid dynamics-based modelling

The most employed fluid dynamics-based modelling technique is the 3D CFD modelling. This modelling technique divides the engine elements into many three-dimensional cells, solving each of them through energy, mass, momentum and state balance equations. This model provides detailed information regarding gas flow, combustion and fuel injection processes in three-dimensional space and this process leads to higher computational power and time [42]. For this reason, it is not recommended to use this type of modelling technique when extensive studies are performed in which thousands of configurations of engine conditions need to be assessed.

2.2.6 Combustion Models in GT-POWER

GT-SUITE is an industry leading simulation software developed by Gamma Technologies to perform engine simulations. It predominantly uses 1D

simulation models combined with several sub programs available for different fields of expertise.

GT-POWER is one of the sub programs of GT-SUITE primarily used to model and simulate internal combustion engines including diesel engines. The program is based on 1D unsteady and nonlinear Navier Stokes Equations.

GT-POWER uses two combustion models which will be explained in detail in the sections below.

- i. Non-predictive Combustion Model
- ii. Predictive Combustion Model

2.2.6.1 Non-predictive Combustion Model

In this combustion model, the burn rate is imposed as a function of crank angle since from start of the simulation setup and will be followed assuming sufficient availability of fuel to support it. For this reason, the burn rate will not be affected by the parameters such as residual fraction or cylinder pressure, and the fuel-air mixture will burn at the prescribed rate [43]. This kind of model can be used to analyse wave dynamics and exhaust configurations, but it cannot accurately predict EGR (Exhaust Gas Recirculation) and injection timings, etc. Examples of such models include study of influence of varying intake manifold length on volumetric efficiency where prediction of burn rate would not be required and the variable of interest has minimal effect on burn rate.

2.2.6.2 Predictive Combustion Model

In this model, the burn rate is first predicted by various parameter inputs such as pressure, temperature, residual fraction, and fuel-air ratio and then further applied in the simulation setup to predict engine performance [43]. This kind of combustion model typically require calibration for various engine parameters to get accurate results. Predictive Combustion models are further divided into single and multizone combustion models and are discussed in detail below.

- i. **Single Zone Combustion Model** - It is a combustion model that consists of only a single overall zone. The cylinder charge here is assumed to be spread uniformly both in composition and temperature during the span of the thermodynamic cycle [44]. The total mixture energy is generated by the enthalpy flux due to fuel injection inside the cylinder and can be calculated simply by using first law of thermodynamics. It is further assumed that the model should be primarily based on empirical heat release laws to predict performance for diesel engines. This makes the computational time of the model fast and

preliminary analysis of the engine performance quite simple [45]. A calibrated single-zone model can accurately perform engine performance and fuel economy analyses but however, they cannot predict changes in the mixture composition and temperatures during the evolution and expansion of the fuel spray in a controlled volume [44].

- ii. **Multi-Zone Combustion Model** - This type of combustion model consists of several burned and unburned zones, each having a separate temperature. The class of multizone model proposed by Hiroyasu et al. divides the spray into radial and axial (penetrating) zones and tracks the evolution of the zones over the time. [44]. The multi-zone models used in GT-POWER are discussed in detail below.

2.2.7 DI Diesel Multi-Pulse Model

It is a multi-zone phenomenological combustion model that is used to estimate the combustion rate and emissions for DI diesel engines with single and multi-pulse injections. The DI multi-pulse model runs faster as compared to the DI Jet combustion model which will be explained later in the section. This model tracks the fuel during all events of fuel injection, evaporation, mixing with the surrounding gas and burning. For this reason, to achieve good results an accurate injection profile and calibration of the model according to the cylinder pressure analysis is necessary [43].

In this model the cylinder volume is divided into three thermodynamic zones having their own composition and temperature. The first zone is the main unburned zone which contains all the cylinder mass at Inlet Valve Closing (IVC). The second zone is called the spray unburned zone which contains the injected fuel and the entrained gas. While the third zone which is spray burned zone contains all combustion products [43]. The four parameters which are primarily used to calibrate this model are Entrainment Rate Multiplier (C_{ent}), Ignition Delay Multiplier (C_{ign}), Premixed Combustion Rate Multiplier (C_{pm}) and Diffusion Combustion Rate Multiplier (C_{df}).

2.2.8 DI Diesel Jet Model

DI Diesel Jet Model is also a multi-zone, multi-pulse phenomenological predictive combustion model developed by Gamma Technologies. In contrast to DI Diesel Pulse Model, the DI Diesel Jet Model divides the injected fuel mass into several axial slices each having five radial zones and has substantially higher computational time as compared to DI Pulse Model and other non-predictive combustion models [43].

Each of these five zones contain additional subzones. As shown in Figure 13, the fuel sub-zone contains the liquid fuel, the unburned sub-zone contains the vapor fuel and other gases such as entrained air, and the burned sub-zone

contains the combustion products. This combustion model only models one nozzle hole so the total mass of the fuel in all zones is the injection rate divided by the number of holes in the actual nozzle. When the first zone travels inside the cylinder, the fuel subzone immediately filled with the liquid fuel and other subzones remain empty. This zone then breaks into droplets which starts to entrain air to evaporate the liquid fuel which forms the second subzone as shown in Figure 13 where it is referred as unburned subzone. The zone starts to decelerate to conserve momentum due to entrained air which causes the outer zones to penetrate less into the cylinder as they entrain air more quickly as compared to inner zones [43].

This unburned zone then starts to combust which is determined by cylinder pressure, temperature and fuel-air ratio inside a zone. The temperature of a zone depends on the injected fuel temperature, entrained air temperature inside a zone and rate of evaporation while zonal fuel-air ratio is dependent on the vapor fuel and air masses in the unburned zone. The unburned fuel and air after combustion move to the burned zone, which further changes the temperature and compositions of that subzone. Emissions such as NO_x and soot particulate are independently calculated in each of the burned subzone [43].

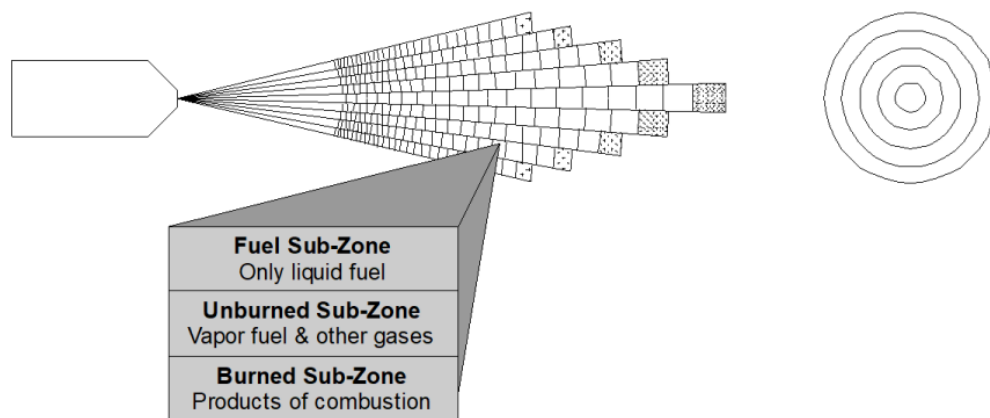


Figure 13: Schematic of the DI Diesel Jet Model [43]

3 Methodology

This section discusses the methodology used to numerically investigate the spray formation and combustion in W25 engine with RCCI combustion configuration. The computational setup is explained, and discussed along with the modelling choices, sub-models, solver configuration, calibration and tuning of the model.

3.1 Computational Setup

The virtual environment of GT-POWER enables the researchers to design, optimize and validate the IC engine models using advanced features of the software such as 1D modelling with comprehensive component libraries and advanced combustion modelling with sensitivity analysis of various parameters. GT-POWER further includes two modules, GT-ISE and GT-POST. GT-ISE is an integrated simulation environment which helps in building, executing and management of the simulation process while the GT-POST is a post processing tool which gives access to the analysis and visualization of the simulated data [43].

The detailed description of the development of the UniNa predictive model is given as follows. An In-cylinder model consisting of a spray jet sub-model, tuned against the constant volume spray chamber experimental results under various ambient conditions, injection timings and rail pressures is integrated with a 0D-multizone model to predict the temporal evolution of fuel distribution inside the cylinder under engine relevant conditions. Furthermore, a reactivity controlled auto-ignition model having Tabulated Kinetic of Ignition (TKI), flame propagation model and emission model are integrated into this 0D multizone model.

This in-cylinder model is then integrated with the 1D commercial code known as GT-SUITE (licensed under Gamma Technologies™) to simulate a W25 single-cylinder research engine fuelled with LFO and Natural Gas. This 1D model is simulated under various engine load sweeps from low load to full load. The methodology used for the validation of the spray and combustion characteristics of the W25 engine using a GT-POWER single cylinder engine model is shown in Figure 14 .

In the following sections, a brief overview of the engine specifications is provided with a description of Three-Pressure Analysis (TPA) model which is used to validate the quality of the experimental pressure trace and calibrate the combustion model to match measured engine performance curves. Running a TPA analysis is helpful to find the inconsistencies between the experimental and simulated air flow rates and valve timings to do relative adjustments before tuning and validating the predictive combustion and spray models integrated with the GT-POWER. The predictive combustion and

spray models will be calibrated and validated by running Design of Experiments (DOE) on various tuning parameters which will be discussed in detail in coming sections and comparing them with the data from the experimental campaigns.

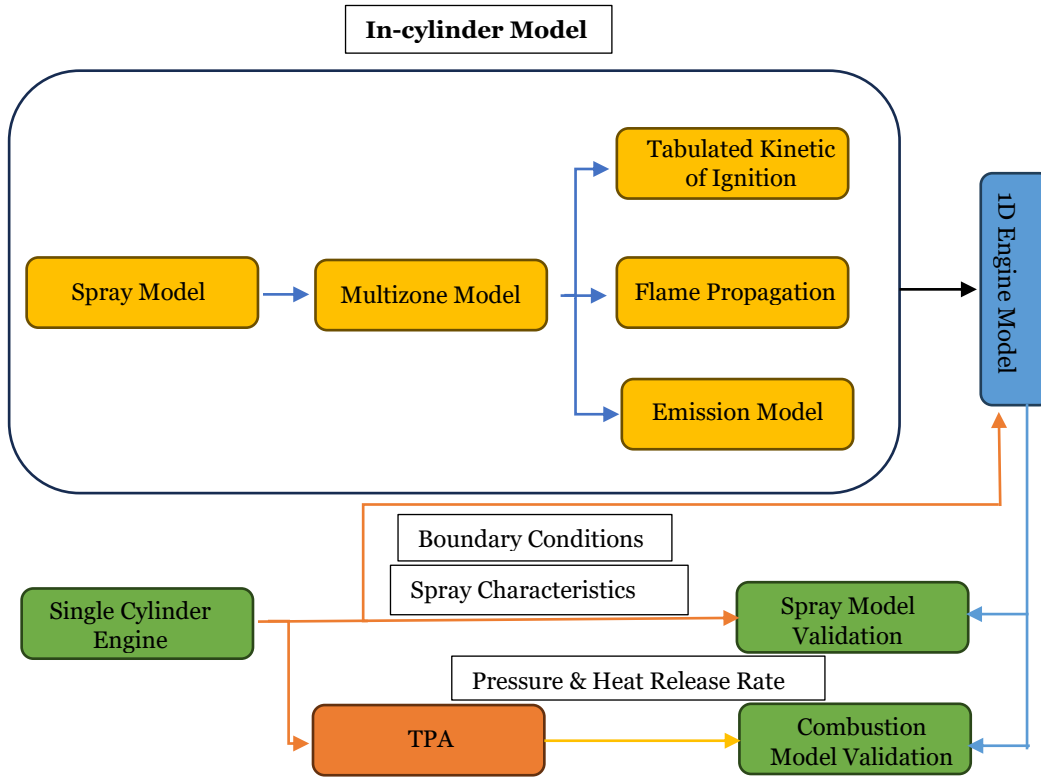


Figure 14: Flowchart for Methodology

3.1.1 Engine Specifications

The Wärtsilä W25 engine is a medium-bore size marine engine designed to operate at medium speeds. The Single Cylinder Engine (SCE) is dedicated for research purposes and can accommodate range of fuels and their combination with advanced technologies. The detailed specification of this engine is outlined in the Table 1. The Wärtsilä W25 medium speed engine is based on the modular design with fuel flexibility ranging from diesel only fuel to dual fuel LNG/Diesel or Ammonia/Diesel. This feature makes this engine one of its kind which can be run on carbon-neutral biofuels and on carbon free ammonia making it to reach beyond today's emission targets.

Table 1: W25 engine and sub-system characteristics

Parameter	Specification
Bore, mm	250
Stroke, mm	340
Nominal Speed, rpm	Variable 750/900/1000
Air System	Pressurized air supply system with temperature control
High Reactivity Fuel System	Common rail direct injection system
Low Reactivity Fuel System	Port fuel gas injection system
Valve Train System	W25DF with Stepless VIC / VEC

3.2 Three Pressure Analysis (TPA)

A Three Pressure Analysis model was developed to calculate the burn rate using measured cylinder pressure trace and validate the quality of measured cylinder pressure by calibrating the engine characteristics such as air-fuel mass flow rates, BMEP, combustion centre etc. against the experimental campaign from single cylinder research engine. As the name suggest this approach requires imposing three pressures in the model; intake, exhaust and in-cylinder, while the UniNa in-cylinder model discussed above is ignored.

The minimum requirement by Gamma Technologies to run the simulation is to have a basic engine model with valve and ports. The simulation will run until the model will converge to steady-state conditions. Two types of analysis are available under this approach: TPA steady and TPA multicycle. As the experimental data available for the W25 engine is assemble averaged so a TPA steady analysis is performed to obtain a single combustion burn rate for each case under study. The cylinder pressure is imposed using EngineBurn-Rate template. Secondary inputs such as air-fuel mass flow rates and emissions are also imposed in the same template while the Combustion Object for Forward Run is also activated with Fuel LHV (Latent Heat of Vaporization) Multiplier turned ON.

To assess the quality of the input data, GT-power performs various consistency checks including followings:

- **IMEP** - The Indicated Mean Effective Pressure (IMEP) obtained from the TPA analysis can be observed in the engine crank train object. This integrated IMEP should be algebraic sum of the Brake Mean

Effective Pressure (BMEP) and Friction Mean Effective Pressure (FMEP). If IMEP has not a reasonable value, then the measured pressure may be inaccurate.

- **Cylinder Pressure at IVC** – During TPA analysis the measured cylinder pressure profile shifts automatically so that a best match of pressure is achieved between the start of the engine cycle and earlier of either 40 degrees BTDC or injection timing. If there is any error, it would mark as an error.

- **Cumulative Burn Rate** - The apparent burn rate during the compression stroke should be close to zero. If the cumulative burn rate is more than 2% of the total fuel it would be flagged as an error.

- **Compression Slope** – The compression slope of the measured LogP vs LogV curve from IVC to start of combustion (SOC) is almost constant and should be close to polytropic co-efficient of trapped gas. If the simulated slope diverges from the measured slope, relevant adjustments should be carried out.

- **Large LHV multiplier change** – The cumulative error in the burn rate calculation is indicated by LHV multiplier. If required changes exceeds 5%, it will be flagged as an error.

- **Air Mass at IVC and Fuel-Air Ratio** - If both measured values are provided in EngCylMeasuredData object, the TPA analysis will compare simulated values and if they differ by greater than 5%, an error will be flagged.

GT-POWER will compare the input data with all above metrics and results can be observed in the cylinder pressure analysis report tables and all consistency checks will be marked with either OK or with the specific error mentioned above. The GT-POWER model used here flagged errors due to inconsistencies in air mass flow rate and compression ratio which was caused by a different sign convention for valve timings used in GT-POWER and relevant adjustments are made as explained below.

3.2.1 Dynamic Valve Timings and Compression Ratio

The GT-POWER simulation environment uses crank angle convention where -180° crank angle is defined as piston's Bottom Dead Centre (BDC) on the compression stroke and 0° crank angle as piston's Top Dead Centre (TDC). While the experimental data from W25 research engine uses -360° crank angle as piston's BDC on the compression stroke. This 180° crank angle difference between the simulated and experimental data introduced inconsistencies during initial TPA analysis as shown in Figure 15. It can be seen from the figure that the peak value of simulated cylinder pressure is less than the measured cylinder pressure peak value and hence lower compression ratio as compared to measured compression ratio.

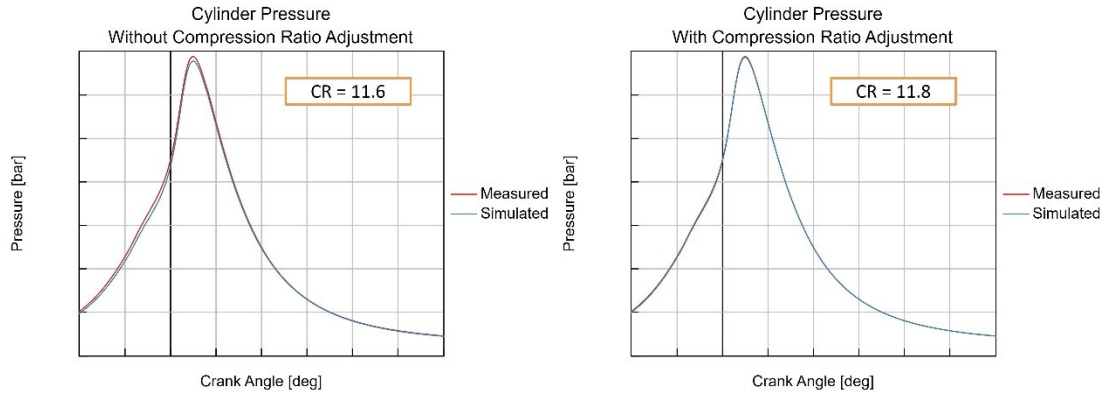


Figure 15: Cylinder pressure trace profiles without (left) and with (right) compression ratio adjustment. Compression ratio left=11.6 and right = 11.8

To fix this problem dynamic valve lifts including dwell timings at maximum lift and a phase shift in crank angle were introduced both at inlet and exhaust valves to get accurate air flow rates and compression ratios. Dwell timings at maximum lift keeps the valve activated until the specific amount of air enters or leaves the engine increasing the volumetric efficiency of the engine.

3.3 UniNa Predictive Model Calibration

After TPA analysis and validating engine characteristics such as air -fuel mass flow rates, maximum cylinder pressure, IMEP and MFB_{50%}, the UniNa multizone predictive model is activated inside the EngCylinder template. The model is run to check for inconsistencies between the measured and simulated data. The tuning of the in-cylinder combustion model is carried out by performing a sensitivity analysis using a total of seven tuning constants. Six of these constants are taken from the diffusion mechanism among the zones ($k_{diff,m}$, $k_{diff,q}$), the spray model ($k_{out,spray}$, So) and from the flame propagation model (c_{trans} , c_{wrk}). The seventh tuning constant taken from spray model is evaporation multiplier ($evap_multi$) which was used to do final tuning after validating spray model. The calibration of the spray jet model is discussed in detail in the thesis in section 3.4. A Design of Experiments for six tuning constants was performed for a total of 7 cases with case 4 as a standard value taken from the phenomenological model presented by Felice and Bellis et al. in [37] for the W31 dual fuel engine fuelled with LFO and Natural Gas. The standard value and the range of tuning constants simulated with the description of their effect on spray evolution and combustion is described in Table 2.

Table 2: Tuning constants description and range of values used for DOE

Tuning Constants		Standard Value	Range
$k_{diff,m}$	controls mass diffusion	0.06	0.015-0.105
$k_{diff,q}$	Controls the heat/thermal diffusion	0.09	0.0225-0.1575
C_{trans}	control the transition from laminar flame to turbulent flame	1.1	0.275-1.925
C_{wrk}	Effects Flame wrinkling	4.1	1.025-7.175
$k_{outspray}$	controls the mass flow exiting zone	8.3	2.075-14.525
s_0	position at which spray expansion starts	0.78	0.195-1.365

The result of the Design of Experiments for the tuning constants can be observed in the figures below. From Figure 17, it can be observed that with the increase in C_{trans} the peak pressure starts to decrease due to delayed turbulence transition from laminar burning process. Other tuning constants show almost no effect on peak pressure except for tuning constant s_0 which until the value of 0.585 shows lower peak pressure values as compared to measured values. These lower values show that no combustion is taking place in the cylinder. With such lower s_0 values the spray leaves the spray core earlier which reduces the fuel concentration in the main combustion zones. Hence the local equivalence ratio is lower to combust the fuel in the main combustion zones. The sensitivity of these tuning constants on combustion phasing is described in following sections.

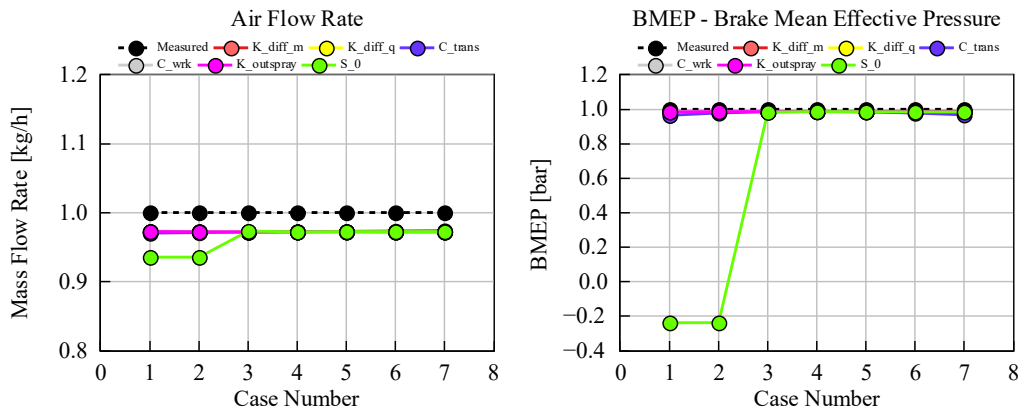


Figure 16: DOE for normalized air flow rate (left) and normalized BMEP (right).

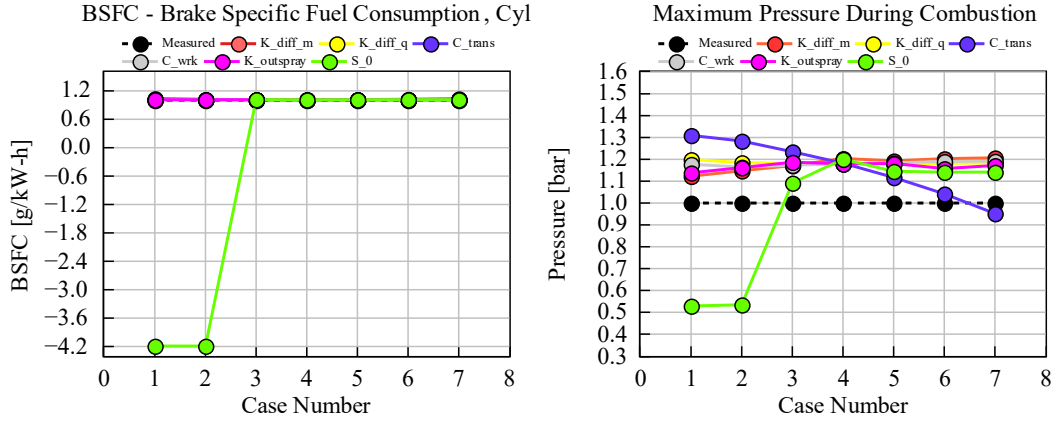


Figure 17: DOE for normalized BSFC (left) and normalized Maximum in-cylinder pressure (right).

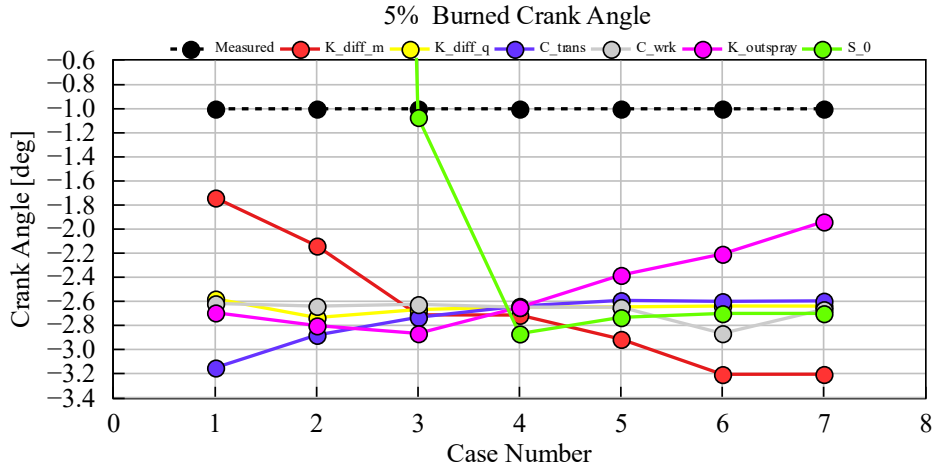


Figure 18: DOE for normalized 5% MFB.

3.3.1 5% Burned Crank Angle Calibration

From Figure 18, the sensitivity analysis for 5% burned crank angle shows that three tuning constants $k_{\text{diff},m}$, $k_{\text{out},\text{spray}}$ and s_0 greatly affect the start of combustion as compared to other constants. So, a set of these tuning constants is fed into design of experiments to tune and calibrate the model against 5% burned crank angle values from the experimental campaign. The validated results for 5% burned crank angle are discussed in detail in the results section 4. As the value of tuning constant $k_{\text{diff},m}$ is increased the start of combustion advances as rate of fuel mass diffusion in the zones far from the injector is increased. On the other hand, higher values of $k_{\text{out},\text{spray}}$ and s_0 are desirable to achieve experimental 5% burned crank angle because as their values are increased the start of combustion retards as higher $k_{\text{out},\text{spray}}$ value leads to leaner mixture in the zone for burning and increasing s_0 delays fuel vaporization.

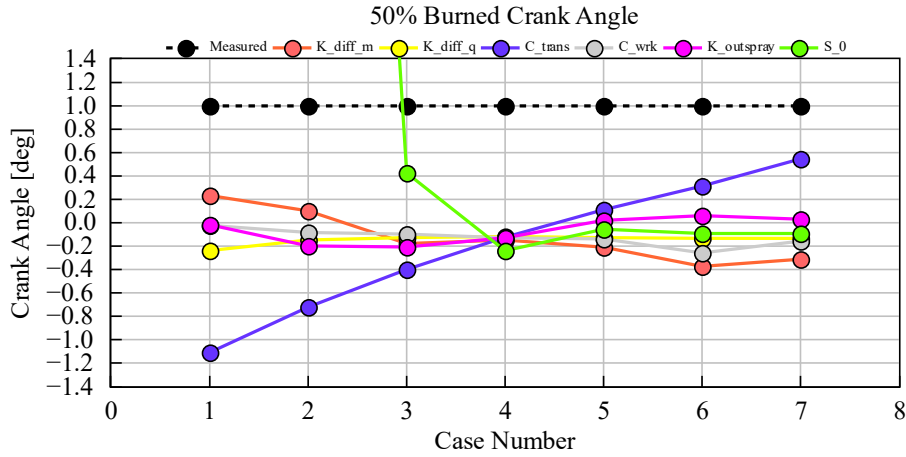


Figure 19: DOE for normalized 50% MFB.

3.3.2 50% Burned Crank Angle Calibration

From Figure 19, it is evident that the 50% burned crank angle is affected by tuning constants taken from flame propagation model i.e. c_{trans} and c_{wrk} . These tuning constants will be used to calibrate the combustion centre for various engine loads. The tuning constant c_{trans} shows unique behaviour as compared to other constant as the combustion centre is delayed almost linearly with the increase in the value of this tuning constant in the range 0.275-1.925. The transition from the laminar burning process to turbulent flame propagation gets slower at high c_{trans} and the transition curve stretches out where the flame stays closer to laminar conditions before it is fully developed into turbulent flame. So, a higher value of c_{trans} will be desirable for calibrating 50% burned crank angle. While by increasing the flame wrinkling factor the turbulent flame speed increases and the 50% burned crank angle advances.

3.4 Spray Jet Model Calibration

After the calibration of the combustion model and validating the engine performance curves, the spray jet model is calibrated. Two tuning constants are used to calibrate the spray jet model, and the results are validated against the measured results from the spray experimental campaign. The experimental values were taken from the cold spray tests under the density of 10kg/m^3 . The first tuning constant used to tune the spray jet model is a spray cone angle multiplier termed as “acono” and the second parameter is evaporation multiplier termed as “evap_multi”. The first parameter mainly effects the calibration of spray jet model while the second parameter will also be used to tune the engine performance curves at final stage.

These tuning constants are used to tune and calibrate three spray characteristics: spray penetration, spray head velocity and spray cone angles. Both tuning constants are altered to an extent until the spray cone angle data measured from experimental campaign matches the simulated data while the spray penetration data will be compared to experiment strictly until 60mm of penetration length which is selected as the best compromise between cold spray testing and in-cylinder evaporative condition while considering typical length of the primary liquid core of the spray jet inside OSCC chamber. The density of fuel spray can be calculated from the model simulations for all load sweep cases at start of injection timings (SOI) using equation of state. The validated results for spray jet model are discussed in detail in the results and discussion chapter.

3.5 UniNa In-cylinder Model Prediction

The UniNa predictive model described above was calibrated against the experimental data with injector configuration $10 \times 0.375 \times 140^\circ$, where 10 is the number of holes, 0.375 is the diameter of single hole in millimetres and 140° is the injector's umbrella angle. Half of this umbrella angle is referred as spray angle. After calibration of both spray jet and combustion models, the model is used to predict the mass of fuel approaching cylinder walls by observing monitor signals for zonal fuel mass inside and outside spray zone. The observations can be stored in a separate GUI file and can be compared with other experimental data sets.

After the model's prediction accuracy was found to a satisfactory level w.r.t experimental data set, it can be further used to predict and compare the mass of liquid fuel approaching on cylinder walls for various operating points spread over the engine operating map. Further simulations were carried out using an experimental data set with injector configuration $10 \times 0.375 \times 164^\circ$, where the umbrella angle used in this experimental setup was 164° . Same as previous experimental data set, 5 cases with varied engine loads were simulated by keeping all tuning constants at same values as before. A comparison for prediction of mass of liquid fuel approaching cylinder walls with both injector configurations was observed using GUI file and the results are discussed in detail in chapter 4.

4 Results and Discussion

4.1 Validation of TPA model

The initial TPA analysis is carried out without integrating the UniNa in-cylinder predictive model to validate the quality of the input data from the experimental campaign. The inconsistencies in air flow rates and compression ratio found during the analysis were fixed using dynamic valve timings and by changing the crank angle convention used in GT-POWER simulation environment. In this way the simulated cylinder pressures matched reasonably with the measured cylinder pressures as described in section 3.2.1. To achieve validation to a satisfactory level relative and absolute error thresholds were used for the engine performance parameters. The relative error threshold for airflow rate, BMEP and BSFC used are 5% while for firing pressure mean it is 10%. For 5% and 50% MFB absolute error threshold of ± 2 CAD is used.

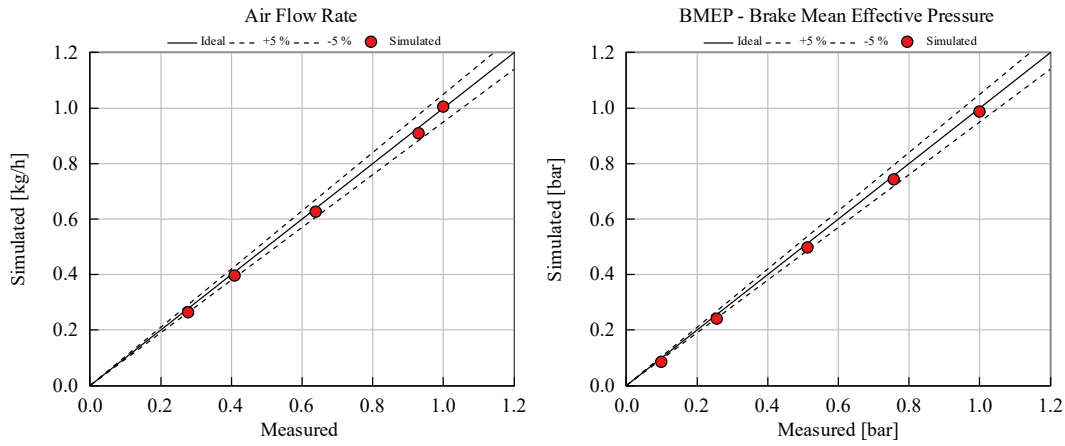


Figure 20: TPA results for normalized air flow rate (left) and BMEP (right).

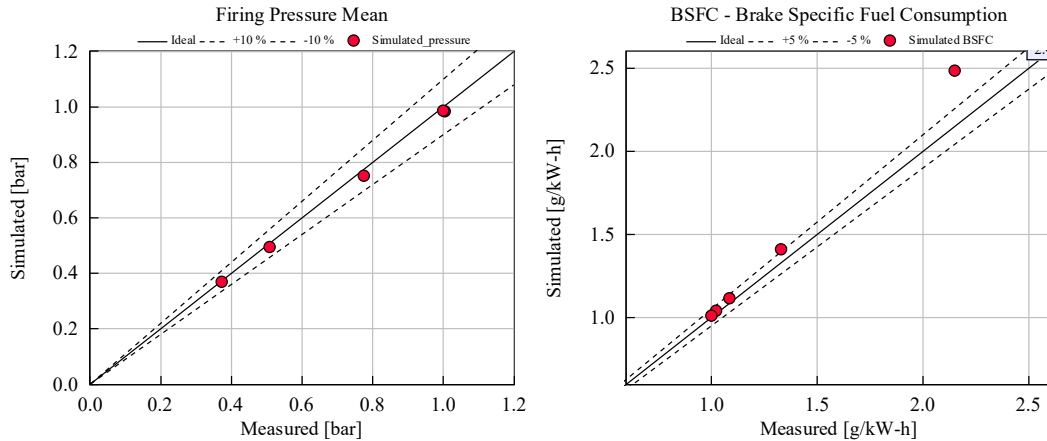


Figure 21: TPA results for normalized Firing Mean Pressure (left) and BSFC (right).

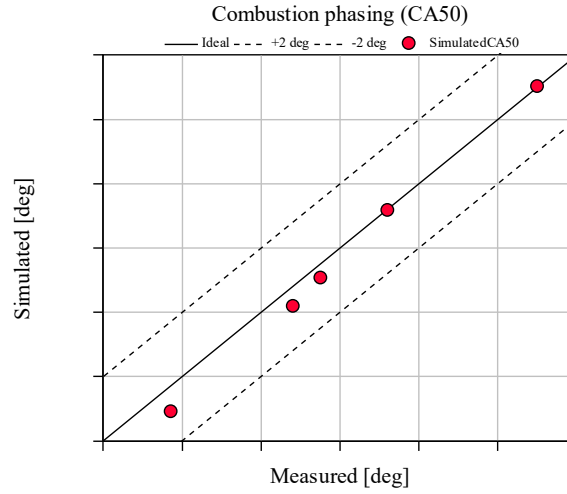


Figure 22: TPA result for 50% MFB.

From Figure 20, the matching of predicted air flow rates and brake mean effective pressure with experimental values can be observed. The simulated data align within the 5% error range which shows that valve timings are effectively calibrated according to the model and indicated mean effective pressure (IMEP) is reasonably comparable to BMEP by a value equal to friction mean effective pressure (FMEP). The TPA results in Figure 21 can be observed for firing mean pressure and brake specific fuel consumption. This simulated data also align within the error range for firing mean pressure values but for lower loads the BSFC graph shows inaccuracies. The reason for this inaccuracy at lower loads can be due to leaner air-fuel mixture at lower loads which are difficult to burn at such lower equivalence ratio and in real engine setups inaccuracies can be introduced due to combustion and heat losses which are usually observed during experimental campaigns in real conditions.

Likewise, the combustion centre (MFB50%) in Figure 22 is also effectively predicted using the TPA analysis. All the values fall within the ± 2 CAD error range which validates the quality of the results from the experimental campaign for combustion characteristics.

4.2 UniNa Predictive Model Validation

The target objective for tuning the UniNa in-cylinder model was to achieve the measured 5% and 50% fuel burned crank angles using series of Design of Experiments during simulation. As discussed in section 3.3, initially six tuning constants explained in Table 2 are used to tune the model. An additional tuning constant “evap_multi” which controls the droplet evaporation during vaporization process was added to further tune and reduce the error between the measured and simulated burned crank angles. To validate the model, a comparison is carried out between the predicted engine performance and the

experimental counter parts as shown in figures below. The model accuracy can be measured by finding the relative and absolute errors between the experimental and numerical data for various engine parameters. The presented data values for all comparisons below are normalized using the measured experimental data at highest nominal load case #5 while the angular data plots for 5% and 50% MFB show only the grid discretization. All the operating conditions shown in figures refer to load sweep with case#1 the lowest and case#5 the highest.

The comparison of experimental and simulated air mass flow rate is illustrated in Figure 23 (on the left). It indicates good level of accuracy between the model and the experimental values with all data points fall within the $\pm 5\%$ error margin. The relative error found between the experimental and measured values is only 2.72% which validates the intake and exhaust pipe geometries, and the valve flow co-efficients specified. The Break Mean Effective pressure illustrated in Figure 23 (on the right) also fall within the error range of $\pm 5\%$ with relative error between modelled and experimental values of just 8.26%. The accurate prediction of both parameters shows that the wall heat losses are reliably estimated.

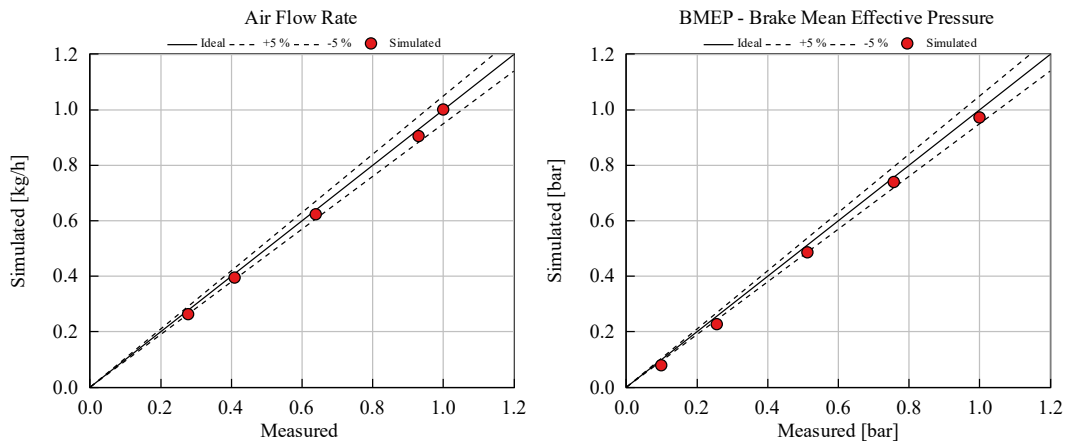


Figure 23: Simulated/experimental comparison of air flow rate and BMEP

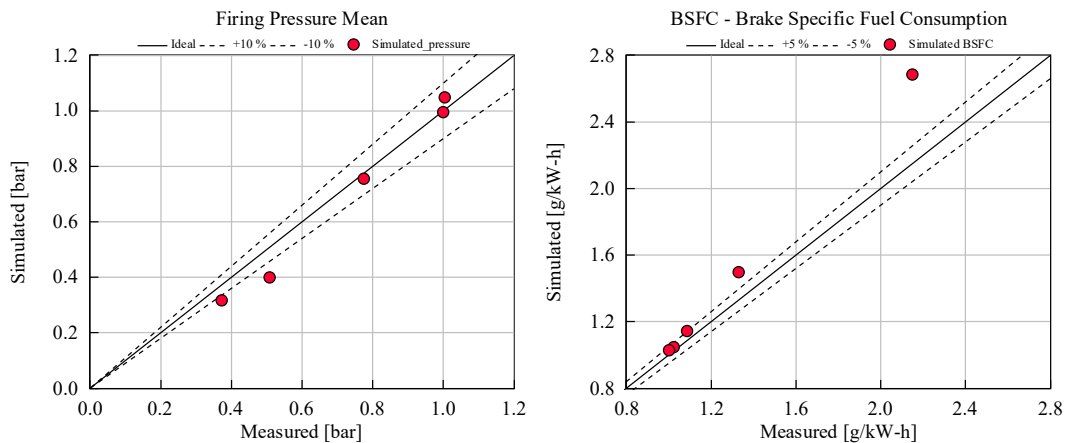


Figure 24: Simulated/experimental comparison of peak pressure and BSFC

The comparison of numerical and experimental peak in-cylinder pressures (on the left) and break specific fuel consumption (on the right) are illustrated in Figure 24. The relative error found between the modelled and experimental data points is 8.76% for peak in-cylinder pressure and 9.59% for BSFC. Two of the five data points lie outside the error margin which are corresponding to the lowest loads case#1 and case#2 which are also responsible for higher relative errors with these engine parameters. But overall, the accuracy of the model is good enough to validate the experimental data for higher loads.

The prediction of start of combustion i.e. 5% MFB (on the left) and combustion centre i.e. 50% MFB (on the right) are illustrated in Figure 25. The absolute errors between the model and experimental values are found to be 1.94 CAD for 5% MFB while for 50% MFB it is 1.58 CAD. Most of the values fall within the ± 2 CAD error band while the values that are outside the error band are for low load cases which have advanced combustion timings. Overall, the combustion timings are well predicted after tuning against combustion timings' objectives which validates the model to a much greater extent.

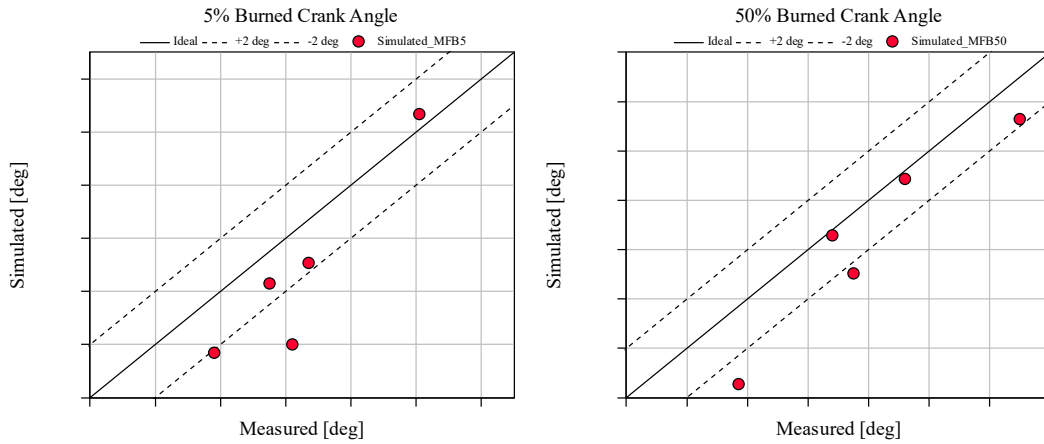


Figure 25: Simulated/experimental comparison of 5% MFB and 50% MFB

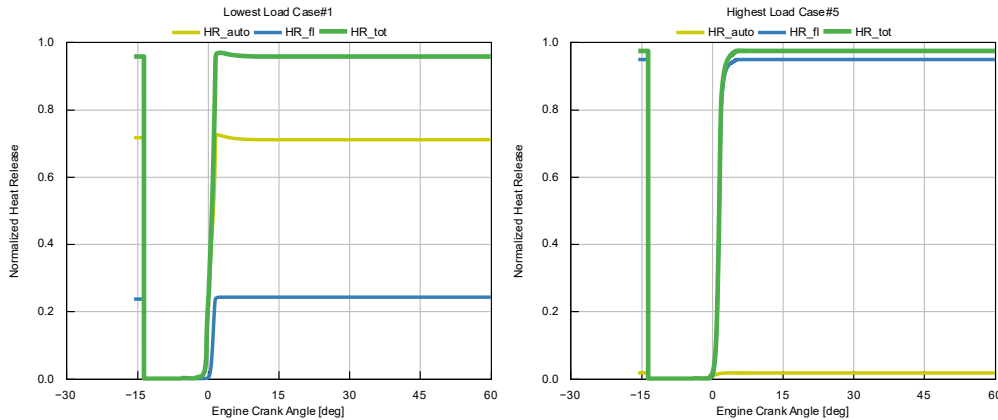


Figure 26: Normalized heat release rates for the lowest and highest load cases

The overall heat release rate for low load case (on the left) and for high load case (on the right) is illustrated in Figure 26. From the former case it is evident that the combustion is primarily governed by auto-ignition (yellow line) while the fuel is distributed uniformly throughout the combustion cylinder. In the latter case the combustion is facilitated by flame propagation (blue line) while auto-ignition (yellow line) occurs in specific zones where fuel is in high concentration levels. The variation of co-existence of both flame propagation and auto-ignition in low and high load cases also validates the combustion model tuned by using combustion or tuning parameters. The final values of the tuning constants found through Design of Experiments and after the model is validated are shown in Table 3.

Table 3: Final Values of the tuning constants used for model

Tuning Parameters	Final Values
$k_{diff,m}$	1.4
$k_{diff,q}$	3.075
c_{trans}	16
c_{wrk}	0.82
$k_{outspray}$	0.05
s_o	0.092025
a_{cono}	0.85
$evap_multi$	1.4

4.3 Spray Jet Model Validation

As explained in section 3.4, two tuning constants are used to calibrate the spray jet model, a_{cono} and $evap_multi$. The latter tuning constant also effect the calibration of the engine performance parameters as discussed in section 4.2, but both constants are mainly used to calibrate and validate the spray jet model. To validate the spray jet model, three spray characteristics from the model simulations are compared with the spray experimental data taken from cold spray test campaign at $10\text{kg}/\text{m}^3$ density. These spray parameters are spray cone angle, spray penetration and spray head velocity. For sake of brevity, the validation results explained below are for only two load sweeps i.e case#1 with 10% load and case#4 with 75% load. The intermediate loads have same trend as illustrated in figures below.

4.3.1 Spray Cone Angles

The spray jet model validation is based on calibrating the spray cone angle until it matches the experimental cone angle values found under cold spray condition at density of $10\text{kg}/\text{m}^3$. The idea to base the validation of spray jet model on spray cone angles is adopted from spray model validation carried

by Felice and Bellis et al. in [46]. A Design of Experiments for parameters “acono” and “evap_multi” was carried out for case#4 with 75% load to find the optimal values where both experimental and model values match each other. The reason to choose case#4 was the density of fuel at SOI which was found to be closer to the cold spray density at which experiments were carried out. The values of the tuning constant found in case#4 were implemented to all other load sweep cases instead of finding them on case-by-case basis. The comparison of model and experimental data points is illustrated in Figure 28 and Figure 28. The modelled values seen in figures matches with the experimental values to a satisfactory level which validates the spray jet model. The difference in experimental and simulated values for case#1 is due to density difference between two data sets.

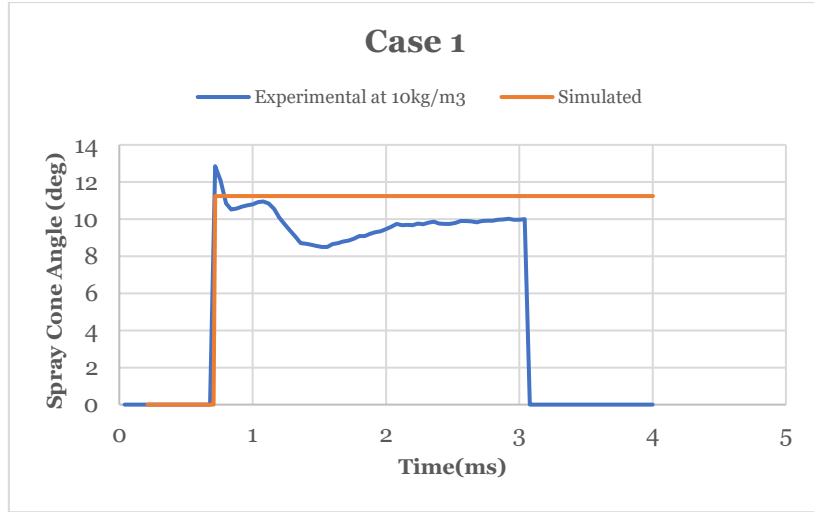


Figure 27: Spray Cone Angle comparison low load case

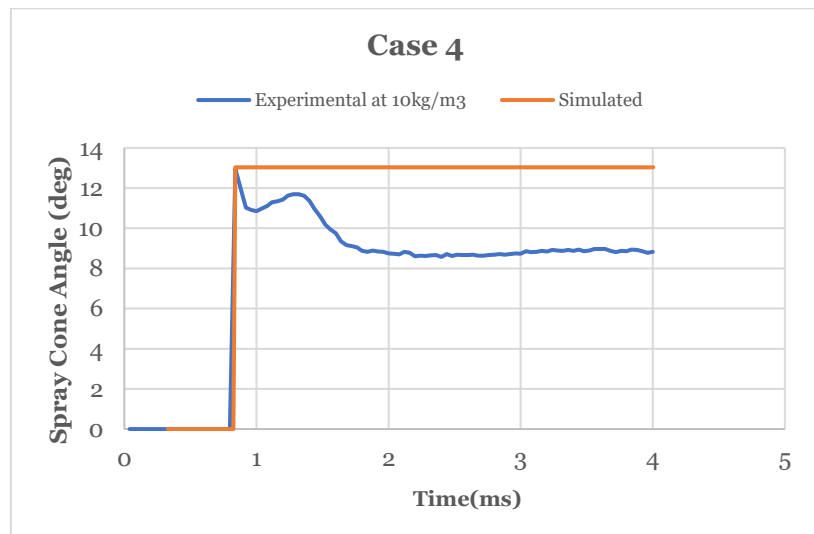


Figure 28: Spray Cone Angle comparison high load case

4.3.2 Spray Penetration

After calibrating the spray jet model based on spray cone angles a comparison between the experimental and simulated spray penetration was observed and illustrated in Figure 29 and Figure 30. From Figure 30, it can be observed that the model can predict the experimental spray penetration trend to a satisfactory level. A penetration length of 60mm is set as a validation threshold, representing a plausible compromise between cold and evaporative pilot-spray conditions, while accounting for primary liquid core length to sustain momentum under similar test conditions. The modelled penetration data matches the penetration trend reasonably at high loads whereas the low load cases show an average error of below 10mm which is apparently due to low ambient density at low loads found at SOI during simulation. This shows the sensitivity of the model to relevant operating parameter such as density.

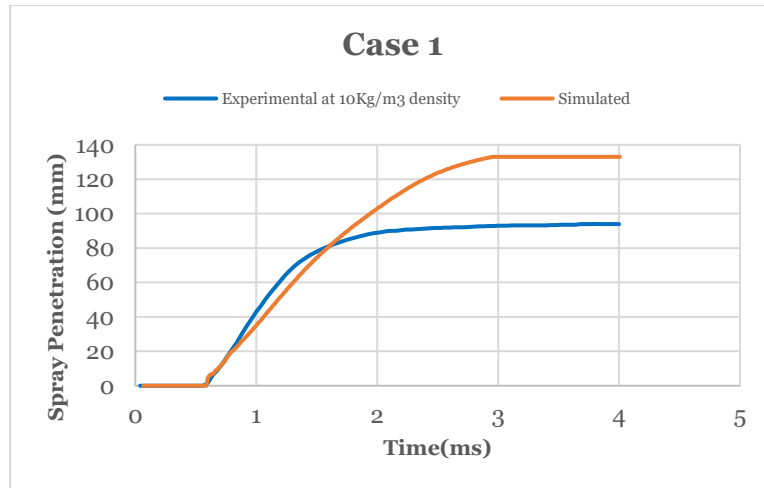


Figure 29: Spray penetration comparison for low load case

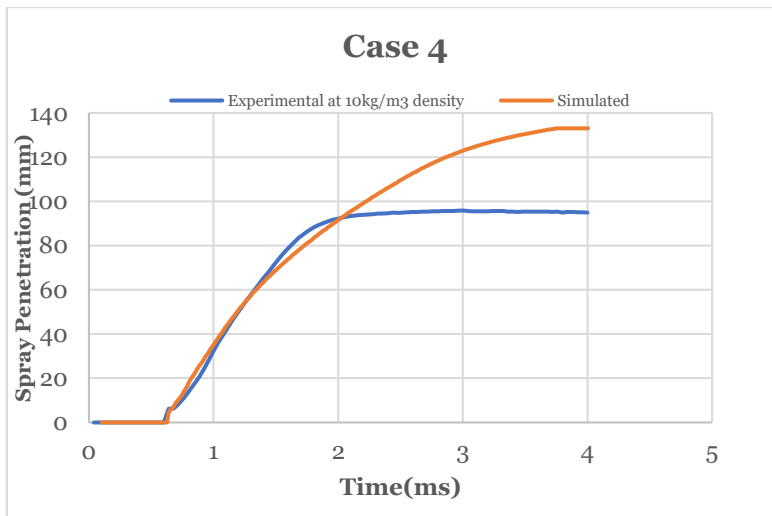


Figure 30: Spray penetration comparison for high load case

4.3.3 Spray Head Velocity

The comparison of spray head velocities at low and high engine loads is illustrated in Figure 31 and Figure 32. The model predicts the spray head velocity to a satisfactory level which is evident from the figures below. The higher spray head velocities during experiments can be due to many reasons. The cavitation inside the injector nozzle can cause higher exit velocities during experiments which may be neglected or simplified during numerical analysis. Moreover, spray breakup and atomization models are based on empirical correlations which may not fully capture the real turbulence, cavitation or secondary breakup of droplets during numerical analysis. The crests and troughs in the simulated data are due to the discretization of the cylinder volume into number of zones where the fuel interacts with the interfacial surfaces of the consecutive zones as fuel progresses farther from the injector.

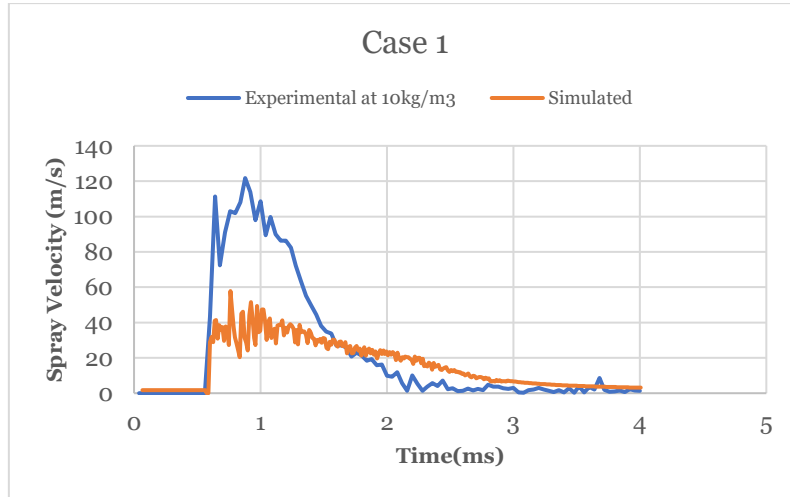


Figure 31: Spray Head Velocity Comparison at low load case

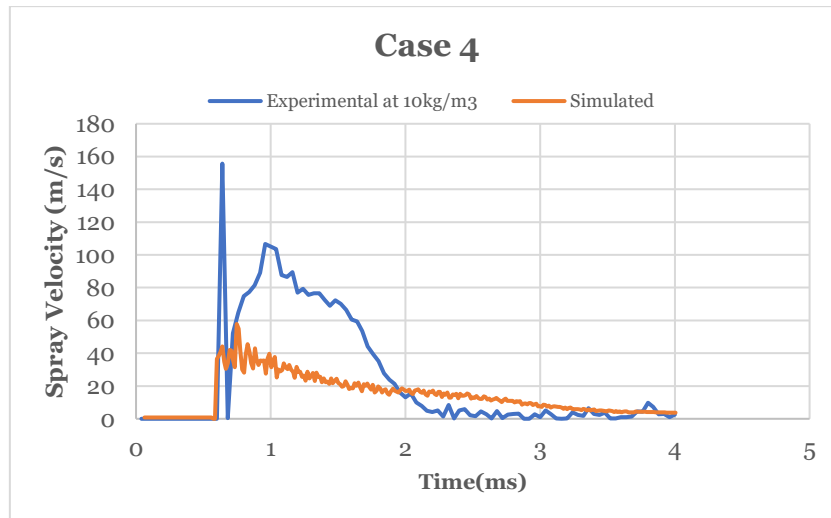


Figure 32: Spray Head Velocity comparison at high load case

4.4 UniNa In-cylinder Model Prediction

As explained in section 3.5, after the model is fully calibrated the fuel mass was predicted near the cylinder walls and analysed using GUI file. Figure 33 presents the amount of liquid fuel in the last zone adjacent to the cylinder walls for a sweep of engine load cases from 10% to 75% at a constant engine speed. It can be observed that cases 1,2 and 3 have large amounts of liquid fuel present near the cylinder walls despite around TDC timing. However, as the load increases from case 2 to case 3 (i.e from 25% to 50%) and from case 3 to case 4 (i.e from 50% to 75%), the liquid mass starts to decrease which means that an increase in the engine load leads to an increase in the vaporization of pilot-fuel spray. The LFO (pilot-fuel) vapor mass can be observed from Figure 33 (left) which shows a greater pilot-fuel vapor mass for case 4 at 75% engine load near the cylinder wall as compared to lower engine loads.

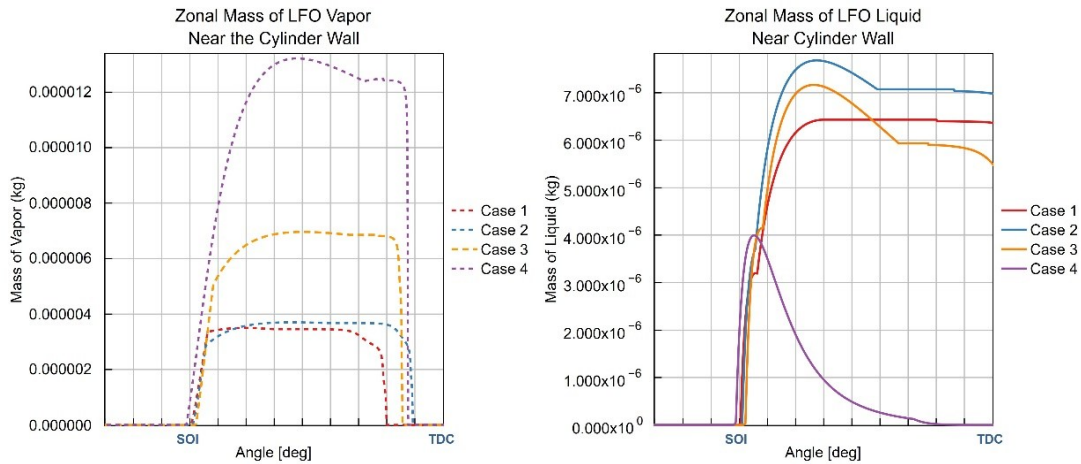


Figure 33: Zonal Mass of LFO Vapor (left) and Zonal Mass of LFO Liquid (right) for load sweep cases 10%, 25%, 50% and 75%

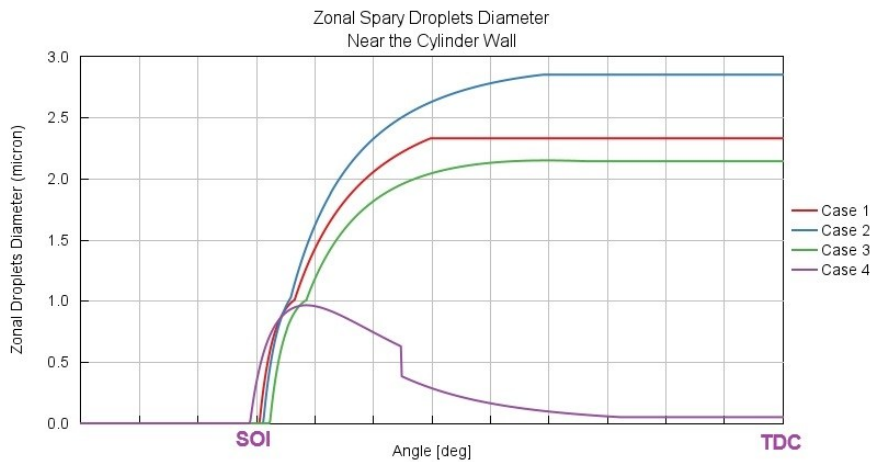


Figure 34: Zonal Spray Droplets Diameter for load sweep cases 10%, 25%, 50% and 75%

The smaller amount of liquid fuel mass and larger amount of vapor mass at higher engine loads can also be verified by observing the spray droplet size behaviour in Figure 34. It is clear from the figure that as the engine load increases from 10% to 25%, initially the droplet diameter increases but increasing the load further from 25% to 50% and from 50% to 75% a clear trend of decreasing droplet diameter can be observed.

Further, to predict the effect of varying nozzle specification such as umbrella angle on the amount of fuel presence near the cylinder wall a comparison of umbrella angle sweep was carried out using two different nozzles. The comparison was performed by analysing the effect of varying umbrella angle on spray head velocity, droplet size and amount of zonal fuel near the cylinder walls. Figure 35, Figure 36 and Figure 37 show a comparison for 50% engine load case. In Figure 35, the comparison of spray head penetration of both liquid and vapor phases for both nozzle configurations can be observed. The simulated 1 refers to the nozzle $10 \times 0.375 \times 140^\circ$ while simulated 2 refers to the injector $10 \times 0.375 \times 164^\circ$ respectively. Spray head penetration in both simulated cases overlap each other without any tangible difference. This indicates that spray head penetration is independent of umbrella angles while number of holes, diameter of holes and discharge pressure remains constant.

In Figure 36, the comparison of fuel spray droplet size for the varying umbrella angles can be observed. It can be seen that the injector with umbrella angle 164° has larger droplet diameter around 3micron near the cylinder wall as compared to the injector with umbrella angle 140° which has droplet diameter of around 2micron near the cylinder wall.

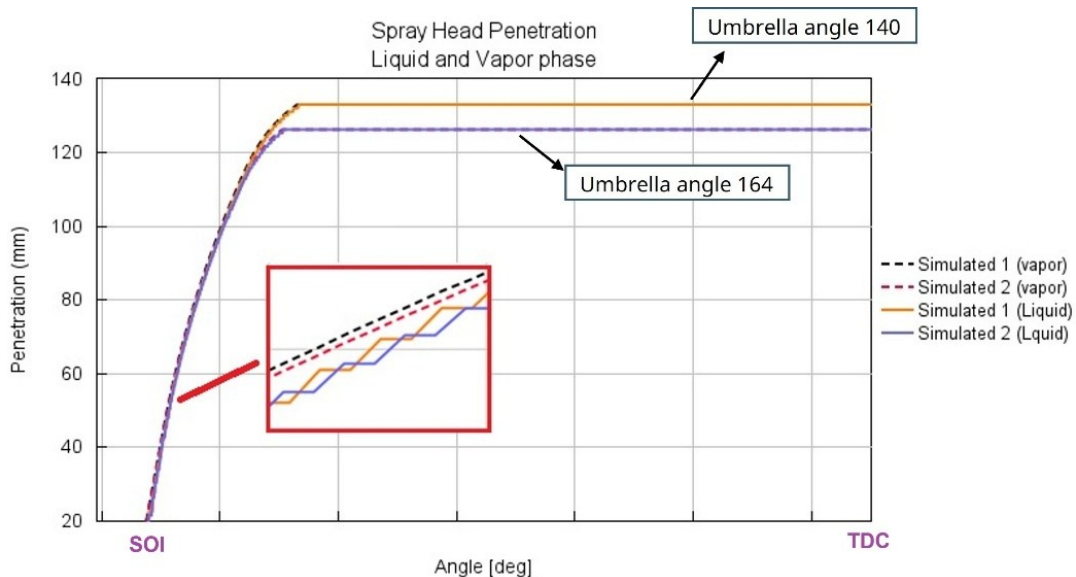


Figure 35: Spray head penetration comparison for sweep of spray angles

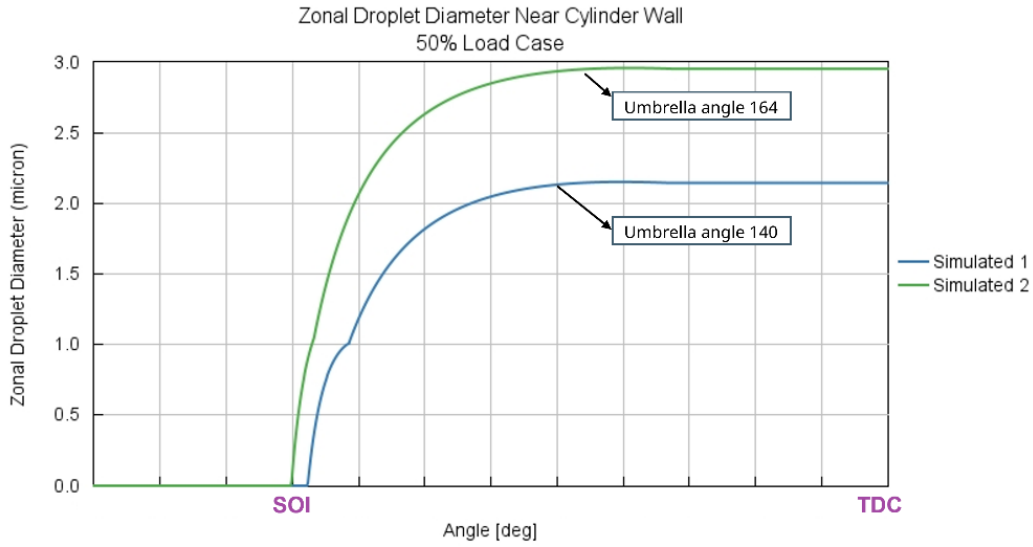


Figure 36: Zonal Droplet Variation for varying injector's umbrella angle (Engine load 50%)

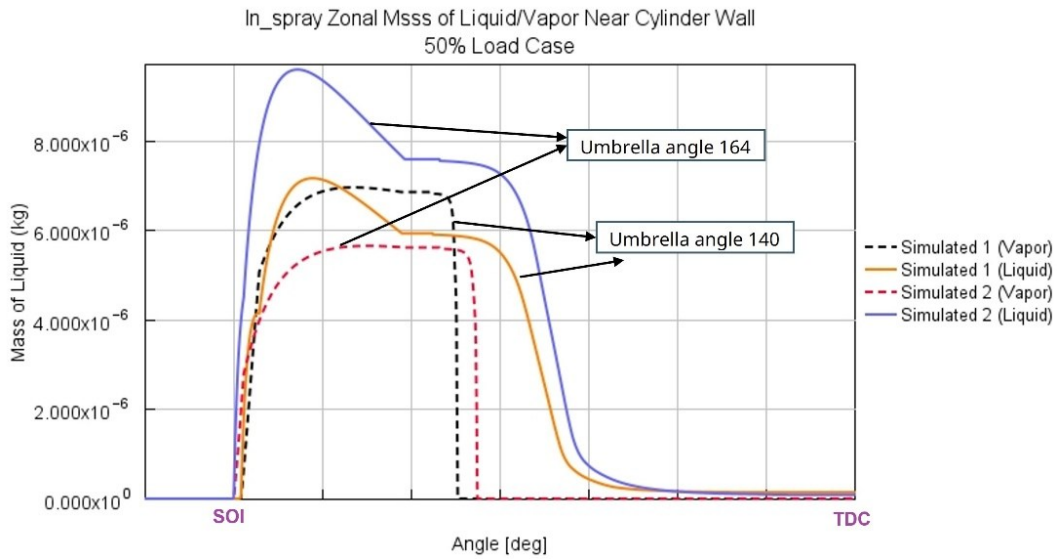


Figure 37: In-spray zonal liquid/vapor mass approaching cylinder walls

The Figure 37 shows the comparison of zonal liquid and vapor fuel masses approaching cylinder walls in the in-spray zone for umbrella angle sweeps. The in-spray zonal region is described in detail in section 2.2.2. The figure clearly indicates higher amount of liquid fuel approaching cylinder walls for simulated 2 case which has larger umbrella angle. With larger umbrella angle (or spray angle), the fuel spray spreads more widely in the cylinder instead of rather staying concentrated towards the piston bowl. The larger the umbrella angle, the radial component of the droplet velocity becomes larger too, leading them to reach the cylinder liner faster.

Moreover, it is seen that both liquid and vapor fuel masses in simulated 1 case with umbrella angle of 140° have almost same amount near the cylinder, but the amount of vapor fuel mass is still higher than in simulated 2 cases with umbrella angle of 164° . This indicates that with smaller umbrella angle (or spray angle) droplets have more time to evaporate before impinging the cylinder walls so eventually less liquid fuel will approach the cylinder walls and may lead to formation of thinner liquid films on the cylinder liner.

4.5 Discussion

As can be observed from sections 4.2 and 4.3 that combustion and spray models calibration to a satisfactory level is necessary to reasonably predict the amount of fuel approaching the cylinder walls. A sensitivity analysis of various engine parameters taken from sub-models such as diffusion model, spray model and flame propagation model was initially carried out and their effect was observed on cylinder peak pressures and combustion phasing. Later, these engine parameters were used to calibrate the 1D UniNa multi-zone predictive model for W25 single cylinder dual fuel research engine to predict the spray evolution and combustion formation inside the cylinder from the zone near injector tip until the zone adjacent to cylinder wall for various engine operating points in a GT-POWER simulation environment.

The presence and evolution of liquid fuel spray in various zones was analysed using GT-POST, a post processing module used in GT-SUITE. The comparison of various injector configurations discussed in section 4.4 shows the effect of variation of umbrella angles on the presence of liquid fuel near cylinder walls. The result matched the expectation of larger amount of fuel presence near the cylinder liner with larger umbrella angle which is due to a change in the geometry of the spray distribution after fuel injection takes place. The geometry changes in such a way that the droplets reach the cylinder walls faster giving less time to evaporate. This also shows that the droplets diameter will also be greater near the cylinder wall with larger umbrella angles as shown in Figure 36 and can cause higher fuel deposition on the wall and consequently produce higher emissions. This droplet behaviour corresponds to the droplet evaporation model given in section 2.2.3 and validates its proper integration with the spray model discussed in section 2.2.1.

The results of combustion and spray models in sections 4.2 and 4.3 shows that some of the tuning constants used for calibration have shown greater sensitivity to the validation results of these models. The tuning constant $k_{diff,m}$ from the diffusion model, c_{trans} from flame propagation model, and s_o and $evap_multi$ from spray model have shown more sensitivity to validation results as compared to other tuning constants under study. Overall, tuning constants from spray model dominated the validation results as they determine the appropriate spray distribution and vaporization of fuel spray droplets which affect the peak pressure and combustion phasing inside the cylinder.

Moreover, from the validation results it can be observed that the lower engine load cases misalign beyond the error margin for some engine performance curves. As explained in the results section, the reason for this deviation can be the operation of lower engine loads on lean mixture conditions which are difficult to burn in real test conditions, and the measured values can therefore deviate from the values found from empirical relations used in simulation environment. The second reason is that the tuning constants cannot replicate their effect on engine performance at higher loads equal to that at lower loads and only one appropriate value is required to represent all load points.

Likewise, in spray model validation results the higher load points replicate the experimental spray characteristics much more satisfactorily as compared to lower loads. During simulation it was found that the low loads have different ambient in-cylinder densities as compared to provided experimental data carried in cold conditions which made them deviate from the experimental data. Therefore, it is recommended in this thesis that the comparison of the simulation results with the experimental results should be at same engine conditions. Moreover, the validated results from both models have overall small relative and absolute errors between the simulated and measured results therefore, it can be used to predict the engine performance and predict liquid fuel presence near the cylinder walls for other operating points spread over the engine operating map. But five operating points used in this validation process may not be able to predict engine performance on other operating points accurately and may have large relative and absolute errors.

However, the validated combustion and spray models can be used further to evaluate the volume of liquid fuel deposited on cylinder walls by extending the wall-film model discussed in section 2.2.4 and validate it using the experimental data that also accounts for zonal fuel approaching the cylinder walls and the temperature variation along the length of the walls. But as discussed above the predictive model should be validated against at least 25 operating points spread over the entire engine map to develop a robust wall-film model which can accurately evaluate cylinder wall-wetting for various engine test conditions.

5 Conclusion and Future Work

This research was aimed at finding the appropriate values of tuning constants shown in Table 3 which were taken from diffusion, spray and flame propagation models to calibrate the predictive in-cylinder model and to predict the amount of fuel approaching the cylinder walls. Based on the simulation results, the model can accurately predict the liquid fuel mass approaching the cylinder liner. It is observed that the liquid fuel mass decrease with the increase in the engine load. The same trend is observed with the droplet size of the pilot fuel spray near the cylinder walls which also decrease as engine load is increased. Further, a comparison of umbrella (or spray) angle sweeps was also carried out to observe the effect of various injector configurations on spray head penetration, droplet size and fuel presence near the cylinder wall.

The comparison indicates that the variation in the umbrella angle has no effect on spray head penetration when number and size of injector holes remain constant. While the variation of umbrella angle does effect on droplet size and fuel presence near the cylinder walls. It indicates that both parameters increase with the increase in umbrella angle of the injectors. Moreover, the tuning constants found after running a series of design of experiments in GT-POWER for all engine load points may not replicate the effect on higher loads equal to that at lower loads and therefore may lead lower load points to deviate from the threshold error limits for the validation results of the combustion model. Lower engine load points in spray model validation also deviate from the experimental data due to difference in ambient conditions considered in both the simulation environment and during experiments.

To make the given model more predictable over the whole engine map, it is recommended to validate the model also for varied EGR rates, injection timings, intake temperatures, pilot quantity and lambda etc. The computation time of the model should also be reduced by using optimal zone discretization between the injector and the cylinder wall to validate the model against parameters mentioned before. Moreover, spray model validation should be based on same ambient conditions used in simulation and during experiments. Lastly, to predict the actual amount of fuel deposition of liquid fuel mass on the cylinder walls, the wall-film model should be extended to account for incoming mass of pilot-fuel and temperature variation along the cylinder height and it should be validated against the experimental data.

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